Why More Agents Do Not Always Help: Empirical Power Laws of Coordination in LLM Agent Societies

Anonymous ACL submission

Abstract

LLM multi-agent systems are increasingly deployed as interacting agent societies, yet scaling often yields diminishing or unstable returns whose cause remains poorly understood. We present the *first large-scale empirical study* of coordination dynamics in LLM multi-agent systems, introducing an atomic event-level formulation that reconstructs reasoning as cascades of coordination. Analyzing over **1.5 million** interactions across tasks, topologies, and scales, we uncover three coupled empirical coordination laws: coordination follows heavy-tailed cascades, concentrates via reinforced routing into influence elites, and produces increasingly large extreme events as system size grows. We show that these regularities are linked by a structural mechanism: an *integration bottleneck*, in which coordination expansion scales with system size while consolidation does not, producing large but weakly integrated reasoning processes. To test this mechanism, we introduce **Deficit-Triggered Integration (DTI)**, which selectively increases integration under imbalance. DTI improves performance precisely where coordination fails, without suppressing large-scale reasoning. Together, our results identify coordination structure as a measurable and previously underexplored axis for understanding and improving scalable multi-agent intelligence. **Code:** https://anonymous.4open.science/r/mas-elites-0645/

1 Introduction

LLM multi-agent systems (MAS) have been widely used in planning, coding, and deliberative reasoning tasks (Chen et al., 2023; Qian et al., 2024; Du et al., 2024; Venkatesh et al., 2026). However, a major challenge lies in their reliability under scaling: adding more agents does not naturally yield proportional gains; instead, performance may plateau, oscillate, or even degrade beyond a certain tipping point (Chen et al., 2024; Cemri et al., 2025). Despite substantial efforts in prompting and reasoning (Wei et al., 2022; Yao et al., 2022), these failures remain stubborn, indicating that the problem is structural and systematic.

These dynamics run deeper than current benchmarks capture. Existing metrics for LLM MAS: task success, final-answer accuracy, and cumulative reward (Liu et al., 2023a), focus on outcomes while overlooking the internal dynamics of collective reasoning. As a result, they cannot distinguish coordinated synthesis from fragmented processes that coincidentally reach correct answers (Zhuge et al., 2024; Cemri et al., 2025), nor capture how effort is distributed, influence accumulates, or coordination stabilizes. Consequently, while prior analyses document scaling failures, they do not explain them systematically (Chen et al., 2024), leaving architectural choices heuristic and interventions reactive (Guo et al., 2024).

What is missing is a science of coordination dynamics; not what agents produce, but how reasoning is initiated, propagated, contested, and resolved (Guo et al., 2024). We show that LLM MAS exhibit cascade-driven dynamics, where agents iteratively respond to prior outputs (Du et al., 2024), and delegation and contradiction induce branching processes. The final output is shaped by the tree of downstream events from an initial reasoning step (Liang et al., 2024). These patterns reveal uneven coordination organized through competing cascades: claims that attract early engagement are more likely to recruit further activity, indicating reinforcement in how reasoning is routed. Coordination is constrained by context length (Packer et al., 2023), communication bandwidth, and token budgets (Gu et al., 2024). Together, these properties motivate a tail-aware, event-level analysis of coordination dynamics in LLM agent societies.

Motivated by these observations, we test whether coordination event sizes in LLM agent societies

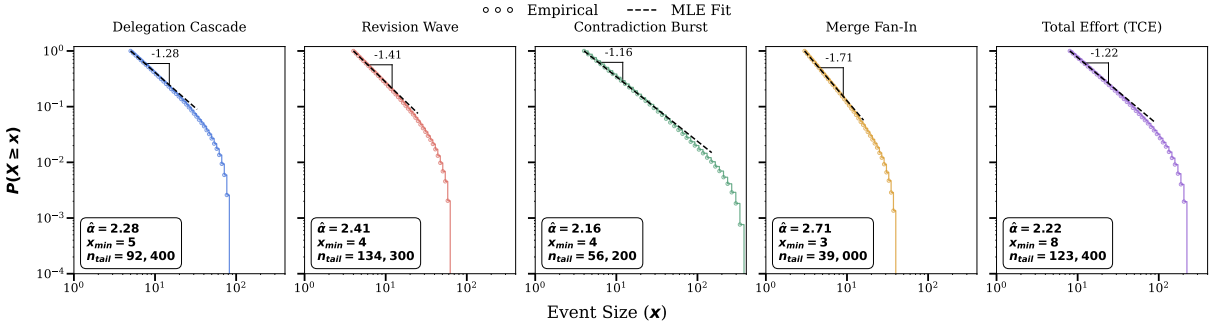


Figure 1: **Heavy-tailed coordination cascades across observables.** CCDFs show broad heavy-tailed regimes with finite truncation. Pooled exponents range from $\hat{\alpha} = 2.16$ to 2.71, and statistical tests support TPL over log-normal and pure power-law alternatives (Table 1). *n_{tail} values are rounded.

exhibit a heavy-tailed structure under finite system constraints (Clauset et al., 2009). We validate this over **1.5 million coordination events** across diverse topologies, tasks, scales, and model families. We derive four coordination-event types: delegation cascades, revision waves, contradiction bursts, and merge fan-in from primitive interaction actions, and define Total Cognitive Effort (TCE) as an aggregate root-cascade measure of downstream coordination. Across pooled observables, event-size tails are well described by truncated power laws with $\hat{\alpha} \in [2.16, 2.71]$, estimated by discrete MLE with KS-selected $x_{\min}$. Run-clustered model comparisons favor TPL over log-normal and pure power-law alternatives after Holm correction ($p < 0.05$). We do not assume power-laws a priori; instead, we statistically test for heavy-tailed structure under finite constraints (Newman, 2005; Stumpf and Porter, 2012). To our knowledge, this provides the first systematic empirical evidence that coordination in LLM MAS exhibits bounded heavy-tailed cascade dynamics, revealing consistent empirical structures with direct implications for how collective reasoning is organized and scaled.

We next examine what these empirical laws imply for coordination at scale. The observed heavy-tailed structure is accompanied by reinforcement in coordination: claims that attract early engagement tend to accumulate disproportionate downstream activity, and this effect strengthens with system size, paralleling preferential attachment in complex networks (Barabási and Albert, 1999). This concentrates cognitive effort in a small subset of agents, forming emergent **`influence elites`**: agents that repeatedly participate in high-engagement cascades and increasingly shape downstream coordination as scale grows. At the same time, the internal composition of large cascades reveals a structural imbalance. As system size grows, expansion driven by delegation and contradiction scales, while integration through merge does not. The system becomes increasingly effective at expanding reasoning without a proportional increase in consolidation, producing an **integration bottleneck** that explains the non-monotonic scaling observed in prior work (Chen et al., 2024; Cemri et al., 2025). Large cascades are not inherently detrimental; they are essential for exploration but become inefficient when misallocated, with a growing fraction reflecting redundant expansion or unresolved conflict rather than effective synthesis.

This finding suggests that improving LLM MAS requires not only increasing reasoning capacity, but also regulating coordination structure. To test this, we introduce **Deficit-Triggered Integration (DTI)**, an intervention that monitors the imbalance between cascade expansion and merge activity and triggers integration when it exceeds a threshold. DTI preserves the heavy-tailed regime while reducing inefficient late-stage expansion. Across topology–task conditions, DTI consistently improves task success, with the largest gains in regimes where the expansion–integration imbalance is most pronounced. These results show that coordination structure is a regulable axis of MAS design, pointing toward a paradigm in which scalability depends not only on model capability, but also on how collective reasoning is organized.

Contributions: (1) We introduce an event-based formulation of multi-agent reasoning that decomposes coordination into atomic primitives and defines Total Cognitive Effort (TCE) to measure downstream reasoning load. (2) We identify three empirical coordination laws: **(H1)** coordination cascades follow truncated power-law distributions, **(H2)** cognitive effort concentrates into emergent influence elites, and **(H3)** extreme coordination events scale with system size. (3) We show

that reinforced routing amplifies already-engaged reasoning trajectories and identify an integration bottleneck in which cascade generation outpaces synthesis, providing a structural explanation for non-monotonic scaling behavior. (4) We propose Deficit-Triggered Integration (DTI), a coordination regulation mechanism that improves performance by balancing expansion and integration.

2 Related Work

LLM Multi-Agent Systems and Coordination: LLM multi-agent systems (MAS) extend reasoning through structured interaction across planning, software development, deliberation, and social simulation (Li et al., 2023; Wu et al., 2024; Hong et al., 2023; Qian et al., 2024; Park et al., 2023). Deliberative mechanisms such as debate and coordination protocols can improve reasoning quality (Du et al., 2024; Liang et al., 2024; Chan et al., 2023), while communication topology strongly influences collective outcomes (Zhuge et al., 2024; Liu et al., 2023b). However, increasing the number of agents does not reliably improve performance and can instead introduce coordination failures, saturation effects, and non-monotonic scaling behavior (Chen et al., 2024; Kim et al., 2025; Cemri et al., 2025). Existing work largely characterizes these failures through end-task outcomes rather than the internal dynamics of coordination itself (Guo et al., 2024; Wang et al., 2024). Our work instead studies coordination as an event-level process and asks whether consistent scaling structure emerges from agent interactions.

Collective Coordination and Influence Concentration: Prior work on collective intelligence and multi-agent coordination suggests that group-level behavior can emerge from local interactions and coordination patterns (Woolley et al., 2010; Lowe et al., 2017; Nisioti et al., 2024). Related studies in social and networked systems show that activity and influence often become unevenly distributed, with a small subset of participants accounting for disproportionate downstream impact (Halfaker et al., 2013; Bakshy et al., 2011; Cha et al., 2010). While these findings motivate the possibility of influence concentration in LLM agent societies, whether such concentration emerges from coordination cascades and how it relates to scaling behavior remains unexplored.

Heavy-Tailed Dynamics and Coordination Cascades: Heavy-tailed distributions and cascade processes arise across many complex systems, including information diffusion, social contagion, and human activity (Barabási and Albert, 1999; Watts, 2002; Leskovec et al., 2007; Barabasi, 2005). Statistical analysis of heavy-tailed behavior requires careful model comparison and inference procedures (Newman, 2005; Clauset et al., 2009; Stumpf and Porter, 2012). We adopt established methodologies for heavy-tail fitting and model comparison, but apply them to a new setting: coordination dynamics in LLM agent societies. To our knowledge, the empirical scaling structure of coordination itself has not been systematically studied.

3 Methodology

3.1 Experimental Setup and Data Generation

We study coordination in LLM MAS using structured workloads derived from four SOTA agent benchmarks: GAIA (Mialon et al., 2023), SWE-bench (Jimenez et al., 2023), REALM-Bench (Geng and Chang, 2025), and MultiAgentBench (Zhu et al., 2025), spanning QA, reasoning, coding, and planning tasks. Each run consists of $N \in \{8, 16, 32, 64, 128, 256, 512\}$ agents solving interdependent tasks under communication topologies including *chain, star, tree, hierarchical, fully connected, sparse mesh, and dynamic reputation*. We adopt a standardized execution protocol: all agents share a common LLM, prompt, tools, and task instances, with execution implemented in LangGraph (Chase and LangChain Inc., 2024) to enforce topology and routing (Sec. J). To maintain balanced coordination demand as N increases, workloads are scaled using a benchmark-conditioned expansion module that generates workload DAGs with sparse dependency structure but no prescribed coordination, allowing interaction patterns to emerge (Appendix I). Agents execute over the workload DAG by iteratively selecting, decomposing, and solving tasks while communicating over the active topology. Interaction traces $\mathcal{T}$ are recorded at the level of individual reasoning steps, capturing agent actions, coordination lineage, and task dependencies, forming the basis for analysis (Appendix F). Experiments are repeated across five seeds per configuration; full details provided in Appendix Section H.

3.2 Event-Based Coordination Formulation

To study coordination dynamics in LLM multi-agent systems in a structured manner, we develop an event-based formulation that decomposes coordination into atomic primitives derived from inter-

action traces. Each trace $\mathcal{T}$ consists of timestamped agent actions with associated references and dependencies, capturing how agents produce outputs and relate them to prior reasoning steps.

We define a *claim* as the atomic unit of reasoning produced by an agent at a given step, corresponding to an intermediate output during task execution. Claims are uniquely identified and linked to prior claims through recorded references, inducing a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{C}, \mathcal{E}_c)$, where $(c_i, c_j) \in \mathcal{E}_c$ if $c_i \in \mathcal{P}(c_j)$. $P(c_j) \subseteq C$ denotes the set of parent claims referenced by c_j. This graph captures the evolution of reasoning across agents, tasks, and interactions. An *event* is a coordination step that transforms or relates one or more claims, identified via the action type and reference structure, capturing primitives such as decomposition, refinement, critique, synthesis, and reuse. Each claim is associated with a root identifier, defining a *cascade* as the set of all claims sharing the same root: $\mathcal{C}_r = \{c_i \in \mathcal{C} \mid \text{root}(c_i) = c_r\}$. Cascades correspond to connected subgraphs of $\mathcal{G}$ representing reasoning propagation from a single claim and form the fundamental units for analyzing coordination dynamics. Detailed notation, definitions of claim types, event categories, and graph construction are provided in Appendix F and B.

3.3 Observables

We define observables over the claim graph $\mathcal{G} = (\mathcal{C}, \mathcal{E}_c)$ and cascades $\{\mathcal{C}_r\}$ to quantify coordination dynamics across multiple scales.

Primitive events and derived observables: From the primitive interaction actions, we construct four coordination-event types for power-law analysis: delegation cascades, revision waves, contradiction bursts, and merge fan-in, whose definitions are provided in Table 17.

Cascade size: The size of a cascade at a root claim c_r is $|\mathcal{C}_r| = \sum_{c_i \in \mathcal{C}_r} 1$.

Total Cognitive Effort (TCE): We define TCE as the total number of downstream coordination events associated with a root claim: $\text{TCE}(c_r) = \sum_{e_k \in E_r} 1$, where $E_r = \{e_k \mid e_k \text{ occurs within cascade } C_r\}$, and E_r contains the downstream coordination events associated with root claim c_r.

Influence concentration: Let $n_a(c_r)$ denote the number of claims produced by agent a in cascade $\mathcal{C}_r$. The top-k influence share is: $S_k(c_r) = \frac{\sum_{a \in \text{Top-}k} n_a(c_r)}{\sum_{a \in \mathcal{A}} n_a(c_r)}$.

Extreme-event scaling: For each coordination observable o, let $\mathcal{X}_{o,j}(N)$ denote the set of observed values in run j at society size N. We define the run-level maximum as $x_{\max}^{(o,j)}(N) = \max_{x \in \mathcal{X}_{o,j}(N)} x$, and report the mean maximum across runs, $\left\langle x_{\max}^{(o)}(N) \right\rangle = \frac{1}{R} \sum_{j=1}^{R} x_{\max}^{(o,j)}(N)$, where o ranges over delegation cascades, revision waves, contradiction bursts, merge fan-in, and TCE.

These observables characterize the distributional structure, concentration, and scaling behavior of coordination in LLM multi-agent systems. In subsequent analysis, we report S_k as E_k^{active} when computed over active agents only, and E_k^{all} when computed over all N agents. Please see Appendix Sec. B for full lists of notation.

Statistical Methods: We analyze coordination observables using heavy-tailed models, including power law, truncated power law, log-normal, and exponential. Parameters are estimated by discrete MLE, with $x_{\min}$ selected by minimizing the Kolmogorov–Smirnov (KS) distance (An, 1933; Smirnov, 1948). Inference uses run-clustered bootstrap resampling, normalized likelihood-ratio tests for TPL–LN, and a boundary-aware nested LRT for TPL-PL. Distributional behavior is visualized using CCDFs on log-log axes (Newman, 2005), following Clauset et al. (Clauset et al., 2009). Additional statistical metrics and reliability tests are provided in Appendix Sec. E.

4 Theory of Reinforced Routing

4.1 Reinforced Coordination Dynamics

Overview: Coordination in LLM multi-agent systems unfolds as a recursive process over claims, where agents iteratively operate on intermediate reasoning outputs through decomposition, refinement, critique, and synthesis. These interactions induce cascades in which claims generate further claims, forming evolving, path-dependent reasoning trajectories: claims that attract more interaction are more likely to be revisited and expanded, while others remain inactive.

We model this behavior as a routing process over claims. Let c_i denote a claim at time t, and let $x_i(t)$ denote its accumulated coordination activity, defined as the number of downstream coordination events involving c_i. Agents select claims non-uniformly, with preference toward those with higher activity. We formalize this using a rein-

forced routing rule:

$$P(c_i \mid \mathcal{F}_t) = \frac{x_i(t)^\beta}{\sum_j x_j(t)^\beta}, \tag{1}$$

where $\mathcal{F}_t$ denotes the interaction history up to time t, $x_i(t)$ is the accumulated coordination activity associated with claim c_i, and $\beta \geq 0$ controls the strength of reinforcement. When $\beta = 0$, routing is uniform over candidate claims; when $\beta > 0$, claims with greater prior activity are preferentially selected, with larger β indicating stronger preferential routing. To characterize this effect empirically, we define the routing ratio $R(x, N)$ as the relative likelihood that a claim with activity x is selected in a system of size N. In the reinforcement regime,

$$R(x, N) \propto x^{\beta(N)} \tag{2}$$

where the preferential attachment exponent is defined as

$$\beta(N) = \frac{d \log R(x, N)}{d \log x}. \tag{3}$$

In practice, $\beta(N)$ is estimated empirically from interaction traces as $\hat{\beta}(N)$, with condition-level estimates denoted $\hat{\beta}$ and event-specific estimates denoted $\hat{\beta}_e$. In isolation, reinforced routing produces broad, scale-free coordination patterns. However, LLM multi-agent systems operate under finite constraints, including limited context, token budgets, communication overhead, and bounded task depth. These constraints suppress unbounded cascade growth, yielding truncated heavy-tailed behavior. We therefore model coordination observables using a truncated power law:

$$P(X = x) \propto x^{-\alpha} e^{-x/x_c} \tag{4}$$

where α governs the tail exponent and x_c captures the finite-size cutoff induced by system constraints such as context length, token budgets, communication overhead, and bounded task depth. Fitted estimates are denoted $\hat{\alpha}$ and $\hat{x}_c$, respectively.

4.2 Cascade Growth and Implications

Under reinforced routing, coordination propagates through claim-rooted cascades that behave as a bounded branching process. Claims that gain early activity are more likely to persist and expand, while others terminate quickly, producing heterogeneous cascade sizes. Finite system constraints cap this growth, yielding broad but truncated distributions. This framework motivates three hypotheses. **(1) Heavy-tailed coordination:** reinforcement produces large variability in event sizes and cascade magnitudes under bounded resources. **(2) Emergence of influence elites:** preferential routing concentrates coordination effort on a subset of claims and the agents contributing to them. **(3) Scaling of extremes:** as system size N increases, the likelihood of large cascades grows, leading to systematic increases in extreme coordination events. We further examine the composition of large cascades and observe an **expansion-integration imbalance**, where expansion primitives (delegation, contradiction, revision) scale more readily than integration primitives (merge/synthesis), producing increasingly large but weakly integrated reasoning structures.

5 Empirical Laws of Collective Cognition

We study coordination in LLM MAS through three hypotheses, evaluated across scales, topologies, task domains, and coordination primitives to uncover underlying mechanisms.

❶ **H1.** Coordination in LLM MAS exhibits bounded heavy-tailed cascade structure under finite system constraints.

❷ **H2.** Collective reasoning exhibits an increasing concentration of coordination effort, producing emergent influence elites whose downstream coordination role strengthens with scale.

❸ **H3.** Extreme coordination cascades grow systematically with agent society size.

5.1 H1: Heavy-Tailed Coordination Cascades

We test H1 by analyzing TCE and its constituent primitives across event types, tasks, topologies, and scales. Figure 1 and Table 1 provide the primary evidence. CCDFs exhibit broad tails followed by systematic truncation. Discrete MLE with KS-selected $x_{\min}$ yields pooled $\hat{\alpha} \in [2.16, 2.71]$; semi-parametric goodness-of-fit tests do not reject the TPL ($p = 0.107$-0.168). Run-clustered model comparisons further favor TPL over LN and PL after Holm correction ($p < 0.05$). This supports bounded heavy-tailed coordination rather than an unbounded scale-free law: most cascades remain local, while a small fraction accumulates disproportionately large activity. This behavior persists with scale (Fig. 2), with $\hat{\alpha}$ stabilizing by $N \approx 64$.

Differences across primitives reveal how coordination processes contribute to cascade growth. Delegation and contradiction exhibit the deepest tails,

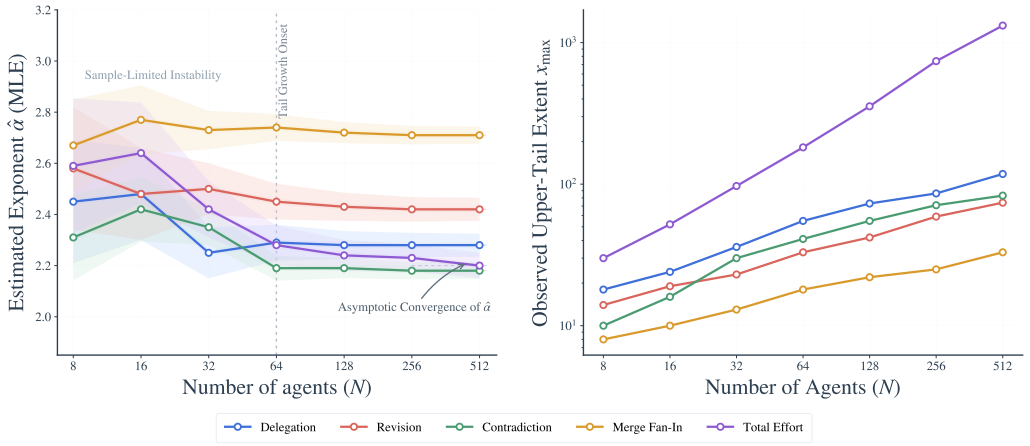


Figure 2: **Finite-size stability of heavy-tailed coordination dynamics. (Left)** Estimated tail exponents $\hat{\alpha}$ (MLE) vs. agent count N. Estimates fluctuate at small N due to limited tail samples, then stabilize and converge beyond $N \approx 64$, indicating emergence of a consistent heavy-tailed regime. **(Right)** Mean maximum event size $\langle x_{\max} \rangle$ vs. N. The upper tail grows systematically across observables, with strongest expansion for TCE, showing that increasing system size expands the reachable coordination tail.

while merge fan-in is sharply truncated (Fig. 1), consistent with their roles: delegation and contradiction introduce new branches, whereas merge consolidates existing trajectories. Cascade growth therefore proceeds mainly through recursive expansion rather than aggregation. This pattern varies with task structure and topology. Tasks with deeper dependency graphs (e.g., planning) support greater recursive expansion, producing heavier delegation and contradiction tails, while merge remains comparatively insensitive (Fig. 5, Table 8). Topology modulates propagation: denser interaction graphs sustain engagement with active trajectories, extending cascades, whereas sparse structures constrain growth. The largest observed cascades occur when deep task structure aligns with high interaction bandwidth; increasing N expands the observable range without altering the bounded heavy-tailed form or TPL preference. Across model families (Table 23), larger models tend to exhibit lower $\hat{\alpha}$ and larger $\hat{x}_c$, extending further into the tail while preserving the same TPL structure. Task- and topology-dependent patterns remain consistent across models (Tables 25 and 24).

H1 is supported: coordination in LLM multi-agent systems exhibits a bounded heavy-tailed cascade structure whose characteristics vary systematically with task decomposability, topology, system scale, model family, and coordination primitives.

5.2 H2: Emergence of Influence Elites

We test **H2** by analyzing how coordination effort distributes across agents and evolves with scale, task structure, and topology. Figure 3 provides the primary evidence. The effort shares of the top-k active agents, E_{10}^{active}, E_{25}^{active}, and E_{50}^{active}, lie well above their egalitarian baselines across all scales, with the gap widening systematically with N (top-10% excess reaching +24pp at large N). The cumulative concentration curves S_p become increasingly convex, indicating that larger systems develop broader and more dominant tiers of **influence elites** rather than converging toward uniform participation. Influence concentration therefore becomes increasingly pronounced with scale.

The observed concentration is accompanied by reinforced routing dynamics in claim selection, evidenced in Fig. 6 and consistent with the reinforced routing model (Eq. (1)) and exponent definition (Eq. (3)). The routing ratio $R(x, N)$ rises above the null baseline with prior engagement and strengthens with N, implying $\hat{\beta} > 0$. This produces a self-reinforcing dynamic in which early-engaged claims attract further delegation and contradiction, forming recursive revision loops that repeatedly draw coordination effort. Delegation and contradiction dominate the high-amplification regime, while merge fan-in remains comparatively local and weakly reinforced. Agents whose claims repeatedly participate in these high-$\hat{\beta}$ trajectories become increasingly central to downstream coordination. Across conditions, $\hat{\beta}$ remains strongly associated with E_{10}^{active} after controlling for society size, task, topology, and model (partial $r = 0.83$, 95% CI $[0.38, 0.94]$), linking local reinforcement dynamics to global concentration.

Figure 9 further provides insight into this scal-

Observable	n_{total} / n_{tail}	$x_{\min}$	$\hat{\alpha}$ [95% CI]	$n_{\text{eff}}/n_{\text{tail}}$	GOF p	TPL–LN: R (Holm p)	TPL–PL: $2\Delta\ell$ (Holm p)	**Stable**
Delegation cascade	342,323 / 92,421	5	2.28 [2.12, 2.39]	0.31	0.143	+3.57 (0.014)	6.73 (0.018)	Yes
Revision wave	344,356 / 134,340	4	2.41 [2.33, 2.67]	0.22	0.132	+2.73 (0.021)	4.51 (0.023)	Yes
Contradiction burst	255,475 / 56,210	4	2.16 [2.13, 2.35]	0.29	0.150	+3.31 (0.017)	5.68 (0.017)	Yes
Merge fan-in	278,533 / 39,128	3	2.71 [2.40, 2.84]	0.13	0.107	+2.33 (0.038)	9.89 (0.029)	Partial
Total Cognitive Effort (TCE)	411,370 / 123,468	8	2.22 [2.11, 2.36]	0.22	0.168	+4.45 (0.013)	6.46 (0.013)	Yes

Table 1: **Global tail fits and statistical validation.** Pooled coordination-event tails are fit using discrete MLE with KS-selected $x_{\min}$. The 95% confidence intervals and model-comparison inference use run-clustered bootstrap resampling of complete executions. GOF denotes the semiparametric goodness-of-fit test for the fitted truncated power law. For the non-nested TPL–LN comparison, $R > 0$ favors TPL; TPL–PL is evaluated using the boundary-aware nested statistic $2\Delta\ell$. Model-comparison p-values are Holm-corrected across the five observables. "Stable" summarizes sensitivity to $x_{\min} \pm 1$; Merge Fan-In shows moderate cutoff sensitivity.

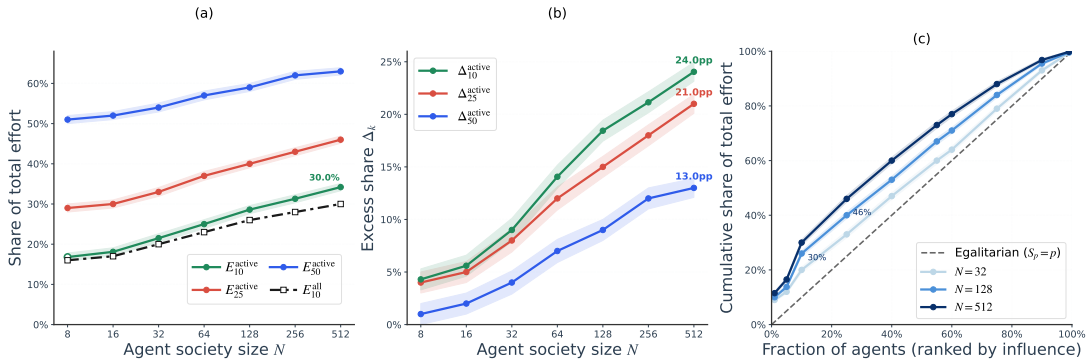


Figure 3: **Scale-dependent emergence of influence elites.** **(a)** Top-k active agents (E_{10}^{active}, E_{25}^{active}, E_{50}^{active}) capture disproportionate shares of coordination effort relative to egalitarian baselines; E_{10}^{all} (dashed) confirms the result is not driven by inactive agents. **(b)** Excess concentration Δ_k^{active} above equal participation increases with N, with strongest gains in the top decile and quartile. **(c)** Cumulative concentration curves vs. N increasingly bow above the equality line, indicating broader and more dominant influence-elite tiers at scale.

ing behavior. As cascade size and N increase, the merge conversion ratio degrades (from 0.21 at small N to 0.07 at $N = 512$ in the top-1% tail), producing increasingly expansion-heavy, merge-poor cascades. Delegation and contradiction compound through recursive revision while integration lags, concentrating unresolved coordination into a smaller subset of agents and amplifying influence concentration. Figure 7 shows the downstream consequence: task success peaks at moderate run-level TCE and drops in the highest TCE quintile, where 68% of runs fail. Failed runs exhibit elevated contradiction and reduced merge shares, indicating accumulation without effective consolidation under expansion-heavy coordination. The association also persists within fixed-N groups (median $r = 0.79$, range $[0.25, 0.89]$), indicating that it is not explained solely by their shared dependence on system scale. H2 is supported: influence concentration increases systematically with scale and is strongly associated with reinforced coordination dynamics, with influence elites playing an increasingly prominent downstream coordination role in larger societies.

5.3 H3: Expansion of Extreme Coordination Cascades

We test **H3** by analyzing how $\langle x_{\max} \rangle$ scales with N across coordination primitives. Figure 8 provides the primary evidence. $\langle x_{\max} \rangle$ increases with N across all observables on log-log axes, with TCE exhibiting the strongest growth ($\hat{\gamma} \approx 0.85$) and merge fan-in the weakest ($\hat{\gamma} \approx 0.42$). Larger agent societies therefore produce not only more coordination activity, but increasingly large cascades, with the observable tail expanding by nearly two orders of magnitude from $N = 8$ to $N = 512$ for TCE.

For reference, an idealized i.i.d. untruncated power law with exponent $\hat{\alpha}$ gives the extreme-value scaling $\gamma_{\text{ref}} = 1/(\hat{\alpha} - 1)$ (Embrechts et al., 2013; De Haan and Ferreira, 2006). The observed TCE exponent ($\hat{\gamma} \approx 0.85$) is close to this finite-range reference ($\gamma_{\text{ref}} \approx 0.82$). Because the empirical cascades are truncated, dependent, and finite-size, we treat this agreement as descriptive rather than as an asymptotic prediction of the fitted TPL.

Differences in $\hat{\gamma}$ across primitives further provide insight into scaling behavior. Delegation and contradiction exhibit stronger scaling than merge, reflecting their association with branching and re-

cursive revision, while merge remains primarily integrative and constrained to consolidating existing branches. As N increases, this asymmetry shifts an increasing fraction of extreme-event mass toward expansion-driven processes. Across model families (Table 23), larger models reach larger $\langle x_{\max} \rangle$ while preserving the same scaling structure.

H3 is supported: extreme coordination cascades grow systematically with society size and exhibit scaling behavior consistent with bounded heavy-tailed coordination dynamics.

6 Law-Aware Intervention: Deficit-Triggered Integration (DTI)

To directly test whether regulating the expansion–integration balance improves coordination outcomes, we introduce *Deficit-Triggered Integration (DTI)*, a targeted intervention that modifies coordination routing under sustained imbalance. DTI operates at the cascade level by monitoring expansion–integration balance (Appendix D); when the integration deficit Δ_r exceeds a condition-specific threshold δ_c, it temporarily prioritizes integration (Fig. 10). Expansion actions are deferred and agents are routed to merge existing branches, increasing merge fan-in and enforcing consolidation. The intervention is local and state-dependent, activated only under imbalance, and does not alter agent capabilities or impose global constraints.

Figure 11 shows that this reallocation produces measurable structural changes. The merge conversion ratio (merge events as a fraction of total expansion events per cascade) increases in high-intensity cascades, reversing the degradation observed in Fig. 9, while contradiction density decreases, indicating improved consolidation of expansion-heavy trajectories. The heavy-tailed distribution of cascade sizes is preserved, showing that large-scale reasoning remains intact while its composition changes. These structural improvements translate into performance gains. Figure 4 shows that gains are largest in high-imbalance settings (Planning×Mesh) and smallest where imbalance is weakest (QA×Chain), a pattern consistent with DTI preferentially benefiting regimes with larger expansion–integration imbalance rather than producing uniform improvements. These results provide interventional evidence that the expansion–integration imbalance contributes to coordination failure. DTI thus provides a proof of principle that coordination structure can be directly regulated. Implementation details and the full algorithm are provided in Appendix D.

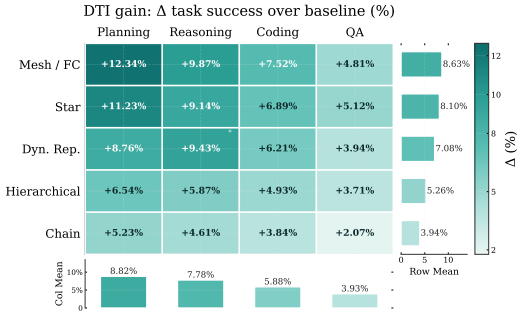


Figure 4: **Heterogeneity of DTI gains across topology and task family.** Relative improvement in task success (%) under DTI versus baseline across topology–task conditions. Gains range from +2.07% (QA × Chain) to +12.34% (Planning × Mesh/FC). DTI produces the largest improvements in conditions exhibiting stronger expansion–integration imbalance. Row and column marginals report topology- and task-level averages.

7 Discussion and Conclusion

Our results show that coordination in LLM MAS follows a consistent empirical structure: bounded heavy-tailed cascades, increasing influence concentration, and systematic growth of extreme coordination events. While these dynamics enable exploration and specialization, they also expose an expansion-integration imbalance: delegation and contradiction scale faster than consolidation, producing large but weakly integrated reasoning trajectories. This offers a structural explanation for why adding more agents does not always improve performance. These findings position coordination structure as a key axis for evaluating and designing LLM MAS beyond model capability or final-task accuracy. Tail behavior, reinforcement dynamics, and integration efficiency provide complementary indicators of coordination health. DTI shows that this structure can be regulated: selectively increasing integration under imbalance improves outcomes without suppressing large-scale reasoning. Scalable LLM agent societies may therefore require not only stronger agents, but mechanisms that shape how collective reasoning is distributed and consolidated.

8 Limitations

Our analysis is limited to a controlled set of coordination primitives, task families, topologies, and agent scales. While the observed patterns are consistent across these settings, longer-horizon deployments and richer coordination mechanisms may exhibit additional dynamics. Our discrete event ab-

stractions enable comparable measurement across conditions, but may omit finer-grained semantic aspects of reasoning quality. DTI is intentionally minimal and targets one failure mode: expansion outpacing integration. It does not address all coordination failures or guarantee optimal performance across regimes. Finally, our results should be interpreted as finite-sample empirical scaling laws rather than universal asymptotic laws; broader theoretical and empirical validation remains important.

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Appendix

A Additional Qualitative Results

In this section, we provide additional qualitative results pertaining to each of the hypotheses tested:

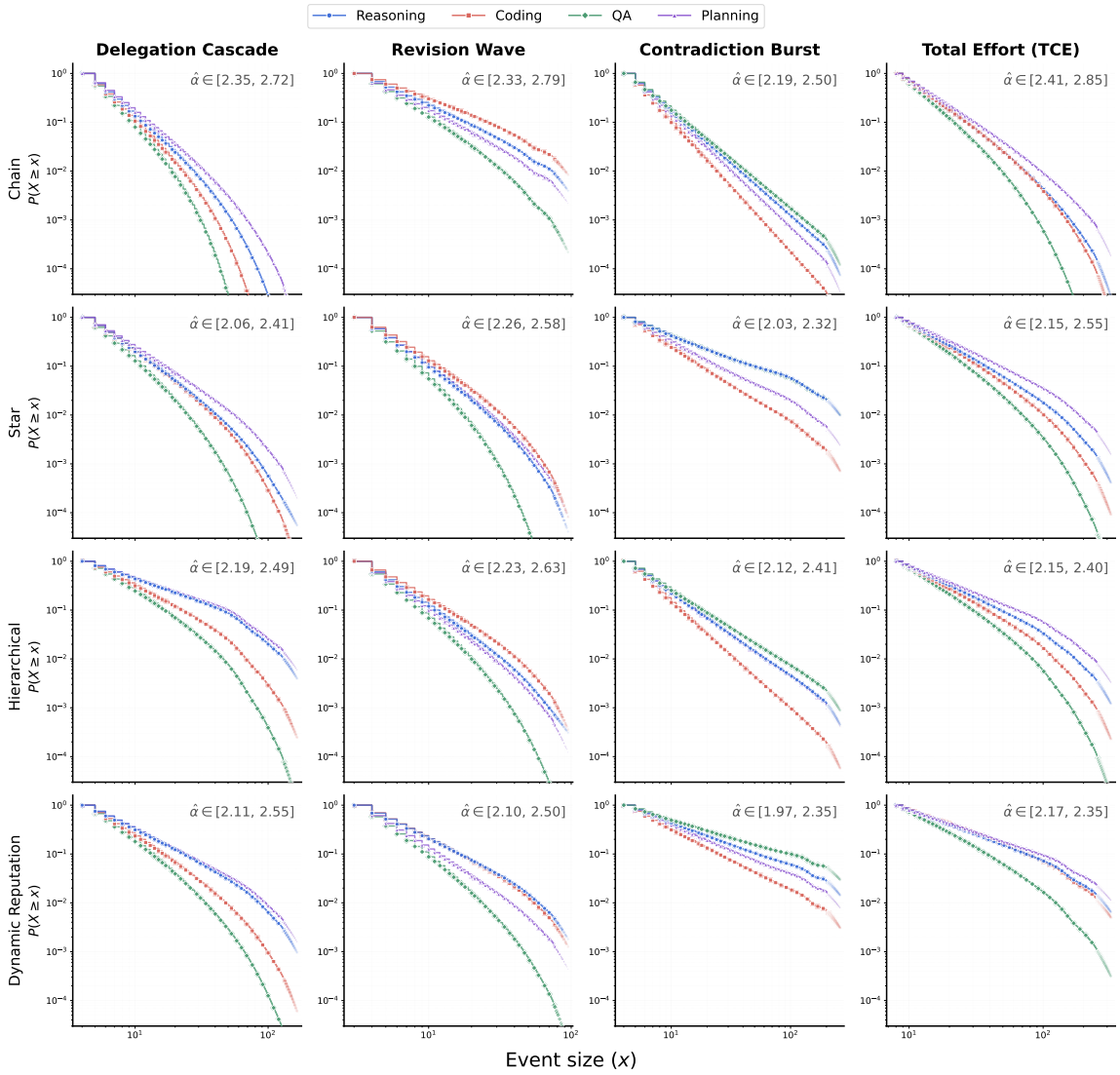


Figure 5: **Topology- and task-specific heavy-tailed coordination cascades in multi-agent LLM systems.** Complementary cumulative distribution functions (CCDFs) of coordination-event sizes $P(X \geq x)$ across four coordination observables: Delegation Cascade, Revision Wave, Contradiction Burst, and Total Cognitive Effort (TCE), under four agent interaction topologies (Chain, Star, Hierarchical, and Dynamic Reputation) and four task families (Reasoning, Coding, QA, and Planning). Each curve represents the empirical distribution of coordination-event sizes for a given task family within a topology. Across evaluated conditions, the distributions exhibit broad heavy-tailed behavior with heterogeneous exponents and finite-size truncation; condition-level estimates are not restricted to $2 < \hat{\alpha} < 3$. While the precise tail exponent varies modestly across tasks and topologies, the heavy-tailed form persists across all evaluated conditions, indicating heterogeneous coordination cascades spanning multiple scales. Deviations at the largest event sizes reflect finite-size truncation due to system constraints such as limited agent attention, bounded communication bandwidth, and task decomposition depth.

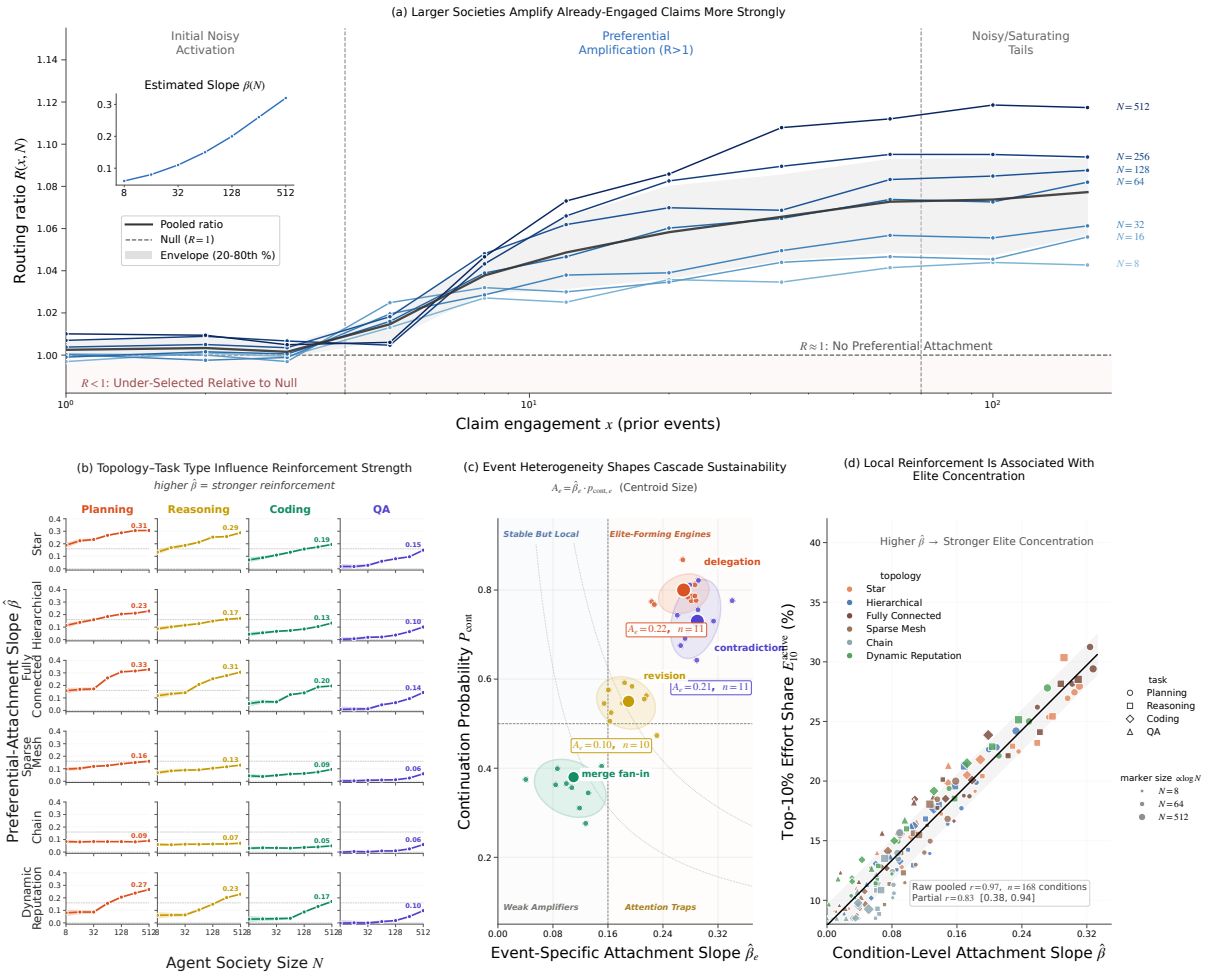


Figure 6: **Preferential attachment and influence concentration in LLM agent societies.** **(a)** The routing ratio $R(x, N)$ rises above the null baseline once a claim accumulates prior engagement, and the effect strengthens with system size N, revealing scale-dependent preferential amplification before saturating in the tail. **(b)** Estimated attachment slopes $\hat{\beta}$ vary systematically across the six topology conditions shown and across task types, with stronger reinforcement in star, fully connected, and dynamic-reputation societies than in chain settings. **(c)** Event types differ in cascade-sustaining power: delegation and contradiction occupy the high-amplification, high-continuation region, while revision is intermediate and merge fan-in remains comparatively local. **(d)** Larger condition-level attachment slopes $\hat{\beta}$ are associated with greater top-10% effort concentration; this relationship remains strong after controlling for society size, task, topology, and model (partial $r = 0.83$, 95% CI $[0.38, 0.94]$).

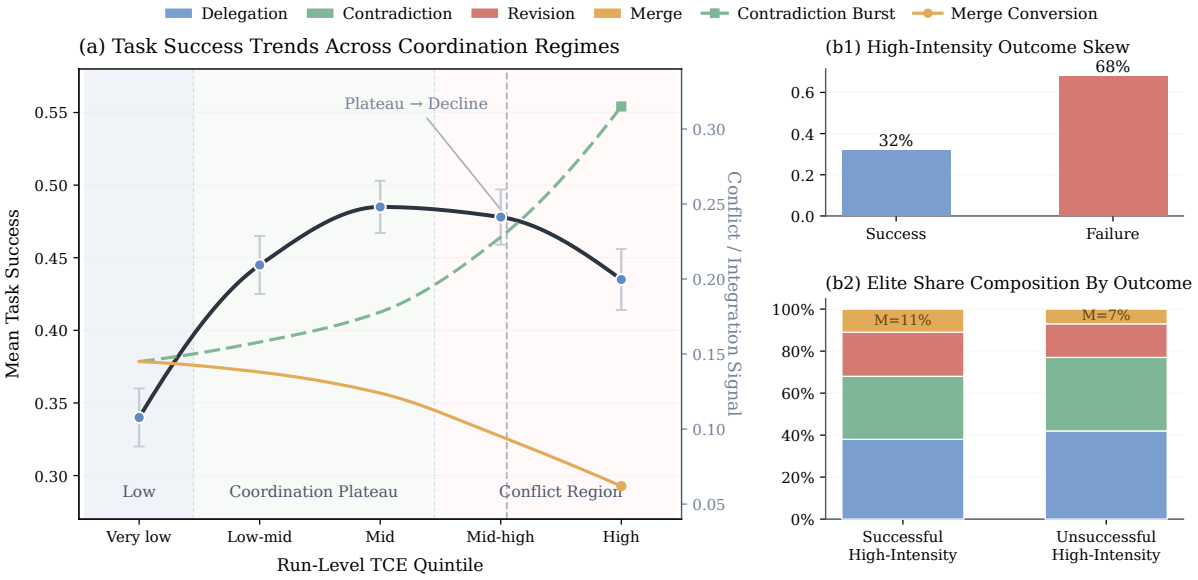


Figure 7: **Conflict-integration dynamics across run-level coordination intensity. (a)** For each completed run, coordination intensity is measured by run-level Total Cognitive Effort (TCE), defined as the aggregate number of downstream coordination events contained in its reconstructed root cascades. Runs are ranked by TCE and divided into five equal-frequency quintiles: 0–20% (*Very low*), 20-40% (*Low-mid*), 40-60% (*Mid*), 60-80% (*Mid-high*), and 80-100% (*High*). The black curve connects the mean task-success rate within these same quintiles; it rises initially, plateaus at moderate TCE, and declines in the highest quintile as contradiction burden rises and merge conversion falls. The black curve and vertical dashed division are descriptive visual summaries, not fitted decision boundaries, optimized thresholds, or causal transitions. **(b1)** Outcome distribution within the highest-TCE quintile, where 68% of runs fail. **(b2)** Influence-elite coordination-event composition for successful versus unsuccessful runs in that quintile; M denotes the merge-event share of influence-elite coordination effort.

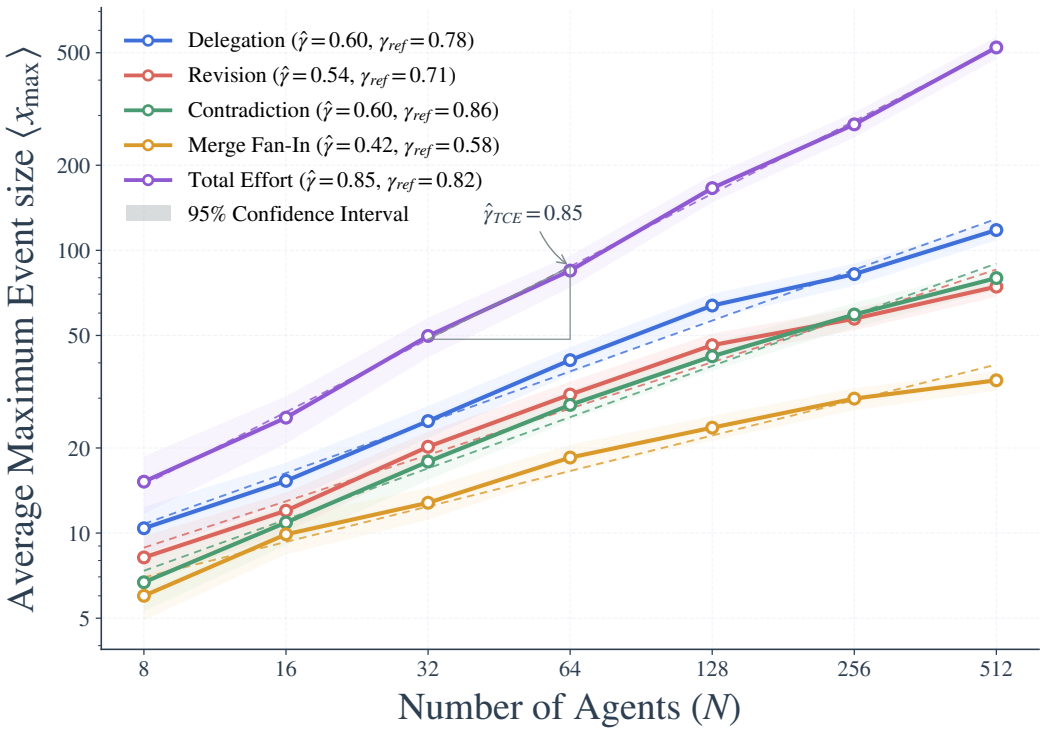


Figure 8: **Extreme-value scaling of coordination observables.** Mean maximum event size $\langle x_{\max} \rangle$ vs. agent count N (log–log). Solid curves show empirical values with 95% CIs; dashed lines denote fitted scaling exponents $\hat{\gamma}$. For comparison, $\gamma_{\text{ref}} = 1/(\hat{\alpha} - 1)$ denotes the idealized i.i.d. untruncated-power-law reference. TCE shows the strongest empirical growth ($\hat{\gamma} \approx 0.85$), close to the corresponding finite-range reference ($\gamma_{\text{ref}} \approx 0.82$).

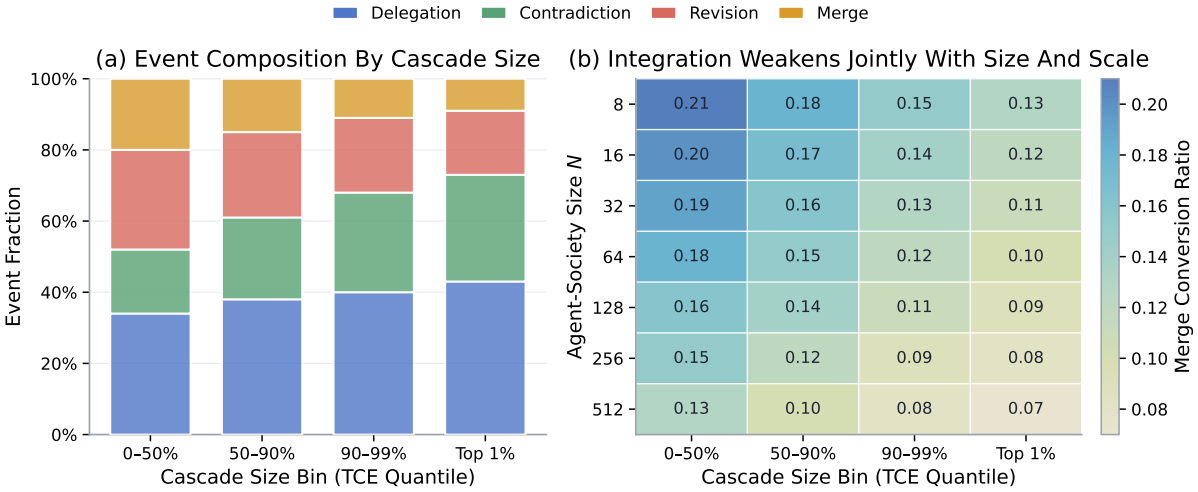


Figure 9: **Internal composition of claim-level coordination cascades and the scale-conditioned integration bottleneck. (a)** Claim-rooted cascades are grouped by total cognitive effort (TCE) quantile, pooling all tasks, topologies, and agent scales. As cascades move into the far tail, their internal event composition shifts toward *delegation* and *contradiction*, while *merge* remains comparatively weak and increasingly subproportional; *revision* plays an intermediate role. **(b)** The same bottleneck appears in the merge conversion ratio across both cascade size and society scale: integration is strongest for smaller cascades in smaller societies and weakens progressively toward extreme cascades in larger societies. Together, these results show that large coordination cascades are increasingly expansion-heavy and merge-poor, and that this integration deficit strengthens with scale.

B Notation

We summarize the mathematical notation used throughout the paper. We use β for the theoretical routing strength and $\hat{\beta}$ for empirically estimated attachment slopes. For H3, we use $\gamma_{\mathrm{ref}} = 1/(\hat{\alpha} - 1)$ only as an idealized untruncated-power-law reference for the observed finite-range scaling.

B.1 Heavy-Tail Modeling

We characterize coordination-event distributions using heavy-tailed statistical models to quantify variability in cascade behavior. The notation below defines the quantities used for tail estimation, model fitting, and likelihood-based comparison across candidate distributions.

B.2 Influence Elite Concentration

We quantify influence-elite concentration by measuring how coordination effort is distributed across agents within each cascade. Top-k shares capture whether a small subset of agents accounts for disproportionate downstream activity, while E_k^{active}, E_k^{all}, and S_p distinguish concentration among participating agents from artifacts of inactive agents.

B.3 Extreme-Event Scaling

We use extreme-event statistics to quantify how the largest observed values of each coordination observable grow with agent society size. The notation below defines the run-level maximum, its mean across runs, and the empirical and reference scaling exponents.

B.4 Intervention (DTI)

We summarize the notation used to describe reinforced routing and Deficit-Triggered Integration (DTI). These quantities define how coordination imbalance is measured within active cascades and how DTI triggers integration when expansion outpaces merge activity.

Symbol	Meaning
X	Random variable representing an event-size observable
x	Realized value of X
$P(X \geq x)$	Complementary cumulative distribution function (CCDF) of X
α	Tail exponent of the power-law / truncated power-law model
$\hat{\alpha}$	MLE estimate of α
$x_{\min}$	Lower cutoff above which tail fitting is performed
x_c	Exponential cutoff scale in the truncated power-law model
$\hat{x}_c$	MLE estimate of x_c
n_{tail}	Number of observations satisfying $x \geq x_{\min}$
LR	Log-likelihood ratio between two candidate models
p	p-value associated with the likelihood-ratio comparison

Table 2: Notation for heavy-tail modeling and model comparison (H1).

Symbol	Meaning
$\text{TCE}(c_r)$	Total Cognitive Effort: total number of downstream events from root claim c_r
$S_k(c_r)$	Top-k influence share within cascade $\mathcal{C}_r$
E_k^{active}	Top-k% effort share; denominator = active agents only
E_k^{all}	Top-k% effort share; denominator = all N agents
Δ_k^{active}	Excess share above egalitarian baseline k/N
S_p	Cumulative concentration curve (Lorenz-style)
$A(N)$	Active-agent fraction at society size N
M	Fraction of influence-elite coordination effort attributed to merge events

Table 3: Notation for influence-elite concentration and inequality (H2).

Symbol	Meaning
$x_{\max}^{(o,j)}(N)$	Maximum observed value of observable o in run j at society size N
$\langle x_{\max}^{(o)}(N) \rangle$	Mean run-level maximum of observable o across runs
$\hat{\gamma}_o$	Empirical extreme-value scaling exponent for observable o
$\gamma_{\text{ref},o} = 1/(\hat{\alpha}_o - 1)$	Idealized i.i.d. untruncated-power-law reference exponent

Table 4: Notation for extreme-event scaling (H3).

Symbol	Meaning
$R(x, N)$	Routing ratio: relative likelihood a claim with activity x is selected
β	Reinforcement strength in the routing rule (Eq. 3)
$\hat{\beta}(N)$	Empirical condition-level attachment slope
$\hat{\beta}_e$	Event-specific attachment slope
$\hat{\beta}_c$	Contradiction scaling exponent used in DTI pressure model
$p_{\text{cont},e}$	Continuation probability per event type
$A_e = \hat{\beta}_e \cdot p_{\text{cont},e}$	Amplification potential per event type
t_r	Coordination events elapsed in active cascade segment
M_r	Realized merge events in current cascade segment
$P_r(t_r)$	Contradiction-driven exploration pressure (Eq. A1)
$\Delta_r(t_r)$	Integration deficit (Eq. A2)
δ_c	Condition-specific deficit threshold (Eq. A3)
a_c	Normalization constant for condition class c
$\mathcal{B}_r$	Active branch heads for root claim r

Table 5: Notation for preferential attachment and Deficit-Triggered Integration (DTI).

C Execution and Compute Accounting

Compute quantity	Value
Complete workflows	∼98,000
Routed interactions / workflow	∼15.3
Total routed interactions	∼1.50M
Maximum API concurrency	50 requests
Open-model hardware	A100-80GB
Total API expenditure	∼$750
Open-model GPU usage	∼600 A100 GPU-hours
Elapsed open-model compute	∼150 h on 4 GPUs

Table 6: **Execution and compute accounting.** Aggregate execution statistics for the experimental campaign. The reported run count denotes complete LangGraph workflows, with model inference performed only by agents selected during routed execution.

The approximately 98,000 runs reported in Table 6 correspond to complete LangGraph workflows, rather than individual model calls. Execution is sparse: only agents selected by the routing policy invoke the underlying model, rather than all N agents at every execution step. Across the experimental campaign, workflows contain approximately 15.3 routed model interactions on average, corresponding to approximately 1.50 million routed interactions overall. API-based experiments were executed asynchronously with up to 50 concurrent requests, while open-model experiments were run on A100-80GB GPUs. Aggregate compute and execution statistics are summarized in Table 6.

D Additional Details on Deficit-Triggered Integration (DTI)

We provide a complete formal specification of Deficit-Triggered Integration (DTI), introduced in Section 6. DTI is a cascade-local intervention that monitors the imbalance between exploration and integration and triggers integration when this imbalance exceeds a condition-specific threshold.

DTI operates at the level of individual coordination cascades. Let r denote a root claim, and consider the cascade defined by all events whose causal ancestry traces to r. For each active cascade, we maintain a local state consisting of the number of coordination events observed so far, denoted by t_r, and the number of realized merge events within the current cascade segment, denoted by M_r. These quantities are updated incrementally as events are generated.

To quantify expansion within the cascade, we model contradiction-driven exploration pressure as

$$P_r(t_r) = a_c \, t_r^{\hat{\beta}_c}, \tag{A1}$$

where $\hat{\beta}_c$ is the empirically observed scaling exponent for contradiction events, and a_c is a normalization constant defined for each condition class c (topology $\times$ task family).

Integration is measured directly through the realized merge count M_r. The difference between exploration pressure and realized integration defines the integration deficit:

$$\Delta_r(t_r) = P_r(t_r) - M_r. \tag{A2}$$

DTI triggers an integration step when the deficit exceeds a condition-specific threshold:

$$\Delta_r(t_r) > \delta_c. \tag{A3}$$

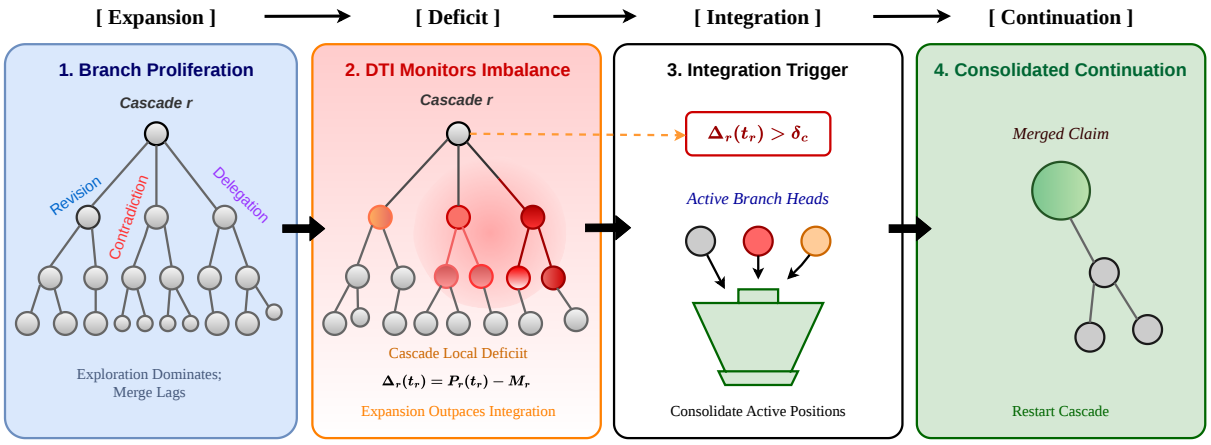


Figure 10: **Deficit-Triggered Integration (DTI) in coordination cascades.** A cascade initially expands through parallel revision, contradiction, and delegation branches, leading to increasing fragmentation (left). DTI monitors this imbalance and triggers integration when it exceeds a threshold, consolidating active branch heads into a unified representation (middle). The cascade then resumes from this integrated state, enabling continued exploration with improved coherence and reduced fragmentation (right).

At this point, the set of active branch heads is collected as

$$\mathcal{B}_r = \text{ActiveBranches}(r), \tag{A4}$$

where $\mathcal{B}_r$ consists of the most recent agent outputs whose causal ancestry traces to root claim r. A structured integration prompt is applied to $\mathcal{B}_r$, consolidating active positions, identifying agreement, and resolving remaining disagreement into a single merged claim. The resulting output is logged as a MERGE event and broadcast as updated shared context.

Following integration, the local cascade segment is restarted by setting

$$t_r \leftarrow 0, \qquad M_r \leftarrow 1, \tag{A5}$$

which reflects that one integration event has already been realized and prevents immediate retriggering.

The parameters a_c and δ_c are estimated directly from baseline coordination traces for each condition class. The normalization a_c captures the empirical relationship between cascade growth and merge activity, while δ_c is defined as the mean plus one standard deviation of the integration deficit observed at cascade termination points. These parameters are estimated directly from baseline coordination traces for each condition class and are fixed prior to intervention, introducing no outcome-tuned quantities.

DTI maintains independent state for each active root claim and therefore requires only $O(|R|)$ additional memory, where $|R|$ is the number of active cascades. Each event incurs constant-time updates to (t_r, M_r), and no additional model calls are made except when the deficit threshold is exceeded.

Because both the normalization and threshold are defined per condition class, the frequency of integration adapts automatically to the underlying coordination regime. Cascades with stronger expansion dynamics accumulate deficit more rapidly and therefore trigger integration more frequently, while cascades with weaker imbalance evolve largely unaffected.

D.1 DTI Intervention Variants

We additionally compare DTI with three intervention variants designed to separate deficit-aware timing from periodic, random, and budget-matched integration.

Control Construction. To avoid introducing separately tuned intervention frequencies, all controls are calibrated to DTI's integration frequency within the corresponding condition. The *fixed-interval* variant invokes the same integration operation periodically, with its interval determined by DTI's mean number of eligible coordination steps per integration. The *random-trigger* variant invokes integration independently at each eligible step with probability equal to DTI's mean trigger rate. The *matched-budget* variant uses the same total number of integration calls as DTI but allocates them independently of the deficit signal. Thus, the controls hold integration frequency or budget approximately fixed while varying whether intervention timing is

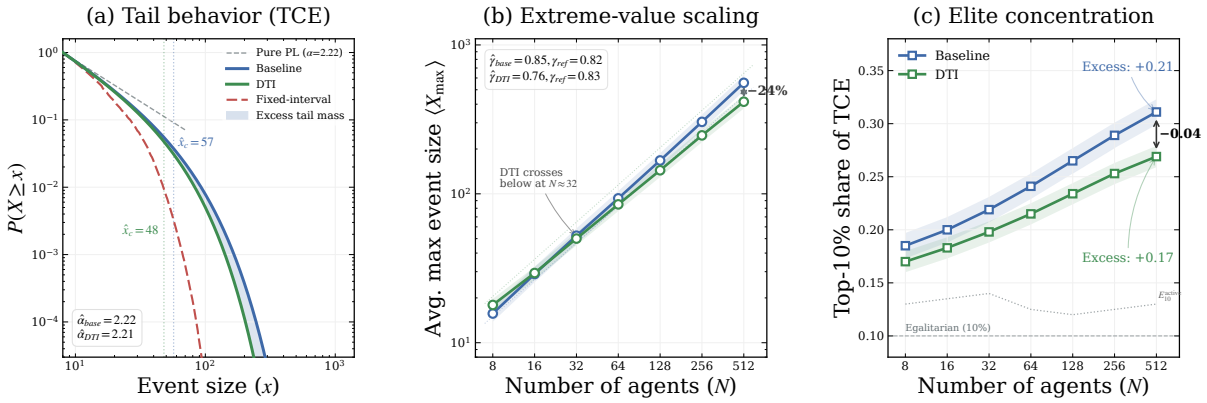


Figure 11: **Impact of Deficit-Triggered Integration (DTI) on collective cognition dynamics.** (a) DTI preserves the heavy-tailed structure of coordination cascades while shifting truncation earlier, reducing excess tail mass without altering the intermediate scaling regime. Fixed-interval intervention, in contrast, introduces premature truncation and distorts the tail. (b) The growth of extreme coordination events with system size is preserved under DTI, but large cascades are systematically attenuated, leading to more controlled scaling behavior. (c) DTI reduces the concentration of cognitive effort among top agents, moderating influence-elite concentration while maintaining a non-uniform and productive coordination structure. DTI converts late-stage expansion into earlier integration, yielding more stable and balanced collective reasoning without disrupting the underlying heavy-tailed organization of agent coordination.

periodic, random, or deficit-aware.

We evaluate these variants at $N \in \{128, 256\}$ in Planning×Mesh and QA×Chain, which represent respectively high- and low-imbalance coordination regimes, where these effects are most pronounced. Tasks, seeds, model, topology, and execution limits are held fixed across variants. Results report task success as mean ± standard deviation over five seeds.

The intervention variants exhibit a clear regime-dependent pattern. In Planning×Mesh, where expansion–integration imbalance is strongest, DTI improves task success over baseline by $+5.05$ and $+7.96$ percentage points at $N = 128$ and $N = 256$, respectively. In QA×Chain, where the measured imbalance is weaker, the corresponding gains are $+0.94$ and $+0.29$ percentage points.

Fixed-interval integration also improves performance in the high-imbalance Planning×Mesh regime, but reduces performance in QA×Chain. Random-trigger integration produces lower task success across the evaluated settings, while the matched-budget variant remains close to or below baseline. Across all four conditions, DTI attains the highest mean task success.

These intervention variants further illustrate the regime dependence of integration: additional synthesis is most useful when expansion substantially outpaces merge-based consolidation, and its effect depends on when integration is introduced. Together with the structural changes reported in Section 6, these results support coordination structure as an additional axis for regulating the scaling behavior of agent societies.

Because each condition uses five seeds, we report effect sizes and dispersion rather than statistical-significance claims.

Method	Planning×Mesh		QA×Chain	
	$N = 128$	$N = 256$	$N = 128$	$N = 256$
Baseline	42.671 ± 1.250	36.145 ± 2.981	53.233 ± 2.865	51.918 ± 2.277
Fixed-interval	44.018 ± 0.876	39.461 ± 1.490	51.280 ± 3.136	49.511 ± 1.620
Random-trigger	39.564 ± 2.303	38.712 ± 2.825	46.201 ± 3.012	41.878 ± 2.112
Matched-budget	41.809 ± 1.652	37.882 ± 1.540	52.198 ± 2.351	48.825 ± 3.655
DTI	$\mathbf{47.721 \pm 2.678}$	$\mathbf{44.105 \pm 1.127}$	$\mathbf{54.177 \pm 2.391}$	$\mathbf{52.207 \pm 0.298}$

Table 7: **DTI intervention variants across coordination regimes.** Task success (%, mean ± SD over five seeds) for DTI and three intervention variants in Planning×Mesh and QA×Chain at $N \in \{128, 256\}$.

Algorithm 1 Deficit-Triggered Integration (DTI)

Require: Event stream $\mathcal{E} = \{e_1, \ldots, e_T\}$; contradiction scaling coefficient $\hat{\beta}_{\mathrm{c}}$; condition-specific normalization a_c and threshold δ_c estimated from baseline logs

For each active root claim r, initialize local cascade state:

$$t_r \leftarrow 0, \qquad M_r \leftarrow 0$$

for $t = 1$ to T **do**
  Observe event e_t
  Determine its root claim $r \leftarrow \mathrm{root}(e_t)$
  Determine its condition class $c \leftarrow \mathrm{cond}(r)$ ▷ topology × task family
  Update local cascade length:

$$t_r \leftarrow t_r + 1$$

  if $\mathrm{type}(e_t) = \textsc{merge}$ **then**
    $M_r \leftarrow M_r + 1$
  end if
  Compute contradiction-driven exploration pressure:

$$P_r \leftarrow a_c \, t_r^{\hat{\beta}_{\mathrm{c}}}$$

  Compute integration deficit:

$$\Delta_r \leftarrow P_r - M_r$$

  if $\Delta_r > \delta_c$ **then**
    $\mathcal{B}_r \leftarrow \mathrm{ActiveBranches}(r)$ ▷ most recent branch-head outputs causally attached to root claim r
    Invoke integration over $\mathcal{B}_r$ to produce merged claim $\tilde{e}$
    Log $\tilde{e}$ as a MERGE event attached to root claim r
    Broadcast $\tilde{e}$ as updated shared context for the cascade rooted at r
    Update merge count:

$$M_r \leftarrow M_r + 1$$

    Restart the local cascade segment:

$$t_r \leftarrow 0, \qquad M_r \leftarrow 1$$

  end if
end for

E Additional Quantitative Results

This section provides additional statistical analyses supporting the empirical laws presented in the main text. Each subsection addresses a specific aspect of robustness, inequality, mechanism, or experimental validity.

E.1 Statistical Validity of Heavy-Tail Fits

Table 1 reports the corrected pooled tail fits using discrete MLE, KS-selected $x_{\min}$, run-clustered bootstrap inference, semiparametric goodness-of-fit tests, and Holm-corrected model comparisons. As a cutoff-sensitivity check, we refit each observable at $x_{\min} \pm 1$. The estimated exponents remain qualitatively stable for delegation, revision, contradiction, and TCE; merge fan-in shows moderate cutoff sensitivity.

E.2 Inequality and Influence Concentration

Beyond top-k share curves, we quantify inequality using the Gini coefficient and effective number of agents. As system size increases, both metrics indicate increasing influence concentration, with a smaller subset of agents accounting for a disproportionate fraction of total coordination activity.

E.3 Tail Anatomy of Coordination Cascades

To understand the structure of large coordination events, we analyze the composition of cascades across size percentiles. Larger cascades are increasingly dominated by expansion dynamics (delegation and contradiction), while merge activity declines sharply, indicating a structural imbalance in integration.

E.4 Task Expansion and Benchmark Accounting

We distinguish benchmark coverage, workload-generation anchors, and random execution seeds throughout the study. The *source pool* denotes the benchmark material available for selecting expansion anchors. *Expansion seed tasks* are benchmark instances used to ground workload generation; these are distinct from the five independent random execution seeds used to repeat each experimental configuration. The expansion procedure then produces benchmark-grounded task instances from these anchors, from which the size-dependent execution workload is constructed as described in Appendix I.

E.5 Event-Definition Sensitivity

We evaluate the stability of the coordination statistics under several reasonable variations in event construction and trace interpretation.

Contradiction-window sensitivity. Contradiction bursts are defined using a temporal aggregation window. We refit the contradiction-burst tail using windows of 1, 2, 3, 5, and 7 steps, with the three-step window corresponding to the primary specification. Across these choices, the fitted tail exponents remain similar and their run-clustered 95% confidence intervals overlap (Table 13). The truncated power-law model remains preferred to the pure power-law and log-normal alternatives under each window.

Root-assignment sensitivity. We additionally compare the primary earliest-ancestor rule for assigning events to root cascades with an alternative construction in which a merge initiates a new root. Under both assignments, the TCE distribution remains adequately described by the truncated power-law model. The corresponding goodness-of-fit values are $p = .168$ and $p = .145$, respectively, indicating that the qualitative tail structure is preserved despite moderate changes in root assignment.

Output-length sensitivity. We also test whether cascade growth primarily reflects variation in generation length rather than coordination structure. Claim length exhibits only a modest raw association with downstream engagement ($\rho = .15$), which falls to $\rho = .009$ with 95% interval $[.005, .017]$ after run-clustered adjustment. Thus, output verbosity contributes little to the measured downstream coordination structure.

		Chain				Star				Hierarchical				Mesh			
Observable	**Task**	$\hat{\alpha}$	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	$\hat{\alpha}$	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	$\hat{\alpha}$	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	$\hat{\alpha}$	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$
Delegation	Coding	2.42	23	+4.1	+2.4	2.29	28	+4.5	+2.3	2.38	26	+4.1	+2.4	2.13	35	+4.4	+2.9
	QA	2.72	17	+3.7	+2.1	2.41	21	+3.9	+2.6	2.49	14	+3.8	+2.3	2.37	23	+4.3	+2.7
	Planning	2.35	34	+4.3	+2.6	2.06	43	+4.2	+3.3	2.19	38	+3.9	+2.8	2.11	57	+5.4	+3.6
	Reasoning	2.37	28	+4.1	+2.6	2.22	35	+4.8	+2.3	2.27	32	+4.2	+2.4	2.09	51	+5.1	+3.5
Merge Fan-In	Coding	2.84	6	+3.4	+3.1	2.77	9	+3.7	+3.3	2.73	7	+3.6	+3.0	2.56	12	+3.3	+3.5
	QA	2.93	4	+2.9	+2.8	2.85	5	+3.4	+3.0	2.89	6	+3.7	+2.7	2.72	9	+3.6	+3.8
	Planning	2.62	11	+3.6	+3.3	2.45	13	+4.3	+3.5	2.58	14	+4.2	+3.2	2.33	19	+4.8	+3.6
	Reasoning	2.74	6	+3.3	+3.2	2.59	13	+3.9	+3.4	2.63	11	+3.5	+3.3	2.41	14	+4.1	+3.7
Revision	Coding	2.56	13	+3.2	+1.4	2.40	18	+3.5	+1.6	2.46	16	+3.4	+1.5	2.31	23	+3.8	+1.7
	QA	2.79	9	+2.3	+1.2	2.58	13	+3.0	+1.3	2.63	11	+2.8	+1.7	2.42	16	+3.7	+1.9
	Planning	2.33	23	+3.7	+1.3	2.27	29	+3.8	+1.7	2.23	25	+4.2	+1.6	2.11	42	+4.3	+2.6
	Reasoning	2.49	19	+3.5	+1.6	2.26	22	+3.5	+1.7	2.35	27	+3.9	+1.5	2.16	31	+4.0	+2.1
Contradiction	Coding	2.31	16	+3.6	+0.6	2.14	22	+3.9	+0.8	2.19	18	+3.7	+0.5	2.05	25	+4.1	+0.8
	QA	2.50	14	+2.6	+0.4	2.32	16	+2.7	+0.7	2.41	12	+3.2	+0.5	2.17	20	+3.9	+0.9
	Planning	2.19	26	+3.7	+0.9	2.03	34	+4.2	+1.1	2.16	32	+4.4	+1.0	1.97	41	+4.6	+1.7
	Reasoning	2.22	21	+3.7	+0.8	2.08	29	+4.3	+0.9	2.12	22	+4.1	+0.9	2.01	33	+4.5	+1.3

Table 8: **Tail heterogeneity across task types and communication topologies.** For each observable and task–topology condition, we report the fitted truncated power-law exponent $\hat{\alpha}$, cutoff parameter $\hat{x}_c$, and model-comparison statistics against log-normal (LN) and pure power-law (PL) alternatives. Each condition is fit independently using discrete MLE with a KS-selected $x_{\min}$, with uncertainty estimated by run-clustered bootstrap resampling; point estimates are shown here for readability. Lower $\hat{\alpha}$ and larger $\hat{x}_c$ indicate broader, less sharply truncated coordination tails. Tail characteristics vary systematically across tasks and topologies, with planning and reasoning generally exhibiting broader tails than coding and QA. Mesh and centralized star communication also tend to support broader coordination tails than chain communication, while planning$\times$mesh yields the largest fitted cutoff scale across the reported observables. For the non-nested TPL–LN comparison, $R_{\mathrm{TPL/LN}} > 0$ favors TPL. For the nested TPL–PL comparison, $2\Delta\ell_{\mathrm{TPL/PL}}$ is reported with boundary-aware inference as described in Sec. 3.3.

N	Top-1	Top-10%	Gini	N_{eff}/N
8	0.04	0.17	0.20	0.88
32	0.07	0.21	0.25	0.81
128	0.08	0.23	0.31	0.73
512	0.11	0.29	0.39	0.64

Table 9: Concentration of reasoning effort across agent scales. Top-k shares increase while the effective number of contributing agents decreases with N.

Metric	Mean	Std (across seeds)
Gini	0.37	0.018
Top-10% share	0.29	0.017
N_{eff}/N	0.64	0.015

Table 10: Stability of inequality metrics across random seeds (averaged over all conditions).

Percentile	Delegation	Contradiction	Merge	Merge ratio
Median	0.30	0.21	0.19	0.37
90th	0.36	0.26	0.15	0.24
99th	0.41	0.31	0.11	0.15
Top 1%	0.45	0.34	0.08	0.10

Table 11: Event composition across cascade size percentiles. Expansion dominates the tail while merge activity diminishes, revealing an integration bottleneck.

Benchmark	Source-pool coverage	Expansion seed tasks	Expanded instances	Avg. dependencies
GAIA	∼150 eligible benchmark tasks	30	150	2.1
SWE-bench Verified	∼235 eligible benchmark tasks	30	150	2.4
REALM-Bench	14 eligible benchmark tasks	11	55	2.0
MultiAgentBench	6 benchmark domains	30	150	2.3
Expansion total	–	**101**	**505**	–

Table 12: **Benchmark and workload-expansion accounting.** Expansion seed tasks are benchmark instances used as anchors for workload generation and are distinct from the five independent random execution seeds used for repeated experimental runs. For MultiAgentBench, the six reported units are benchmark domains rather than six individual task instances; five anchors are sampled per domain, yielding 30 expansion seed tasks. The source-pool column is therefore descriptive and is not summed across benchmarks because its units differ across benchmark suites.

Window	n_{tail}	$\hat{\alpha}$ [95% CI]	TPL preferred
1	39,673	$2.21\,[2.12, 2.43]$	Yes
2	43,286	$2.21\,[2.10, 2.38]$	Yes
3 (paper)	56,210	$2.16\,[2.13, 2.35]$	Yes
5	53,120	$2.17\,[2.09, 2.31]$	Yes
7	49,892	$2.19\,[2.16, 2.29]$	Yes

Table 13: **Sensitivity of contradiction-burst tails to the temporal aggregation window.** Tail estimates remain similar across window choices, with overlapping run-clustered 95% confidence intervals.

Root rule	$x_{\min}$	n_{tail}	$\hat{\alpha}$ [95% CI]	GOF p
Earliest ancestor	8	123,468	2.22 [2.11, 2.36]	.168
Merge starts new root	9	108,100	2.43 [2.09, 2.57]	.145

Table 14: **Sensitivity of TCE tails to root assignment.** The earliest-ancestor rule corresponds to the primary TCE specification in Table 1. The truncated power-law description remains adequate under both root constructions.

F Event-Level Coordination Formulation and Trace Construction

F.1 Coordination Hierarchy

We distinguish four levels of structure in multi-agent reasoning:

- **Task:** the global problem instance to be solved.
- **Subtask:** a decomposed work unit created via delegation.
- **Claim:** a unit of reasoning, such as a proposed solution, critique, or intermediate result.
- **Event:** a single coordination step that creates, modifies, routes, or combines claims and subtasks.

This hierarchy separates *what is being solved* (tasks and subtasks) from *how reasoning evolves* (events acting on claims). Our analysis operates at the event level: all coordination observables are defined over collections of events rather than over entire tasks. Importantly, claims represent reasoning artifacts, while events represent transitions between them.

F.2 Claim and Event Definitions

We formalize claims and events as the fundamental elements of coordination dynamics.

Claims: A claim represents a unit of reasoning produced by an agent, such as a proposed solution, refinement, critique, or synthesis. Each claim is assigned a unique identifier and may reference one or more parent claims, forming a directed lineage structure.

Formally, a claim c_i is associated with a set of parent claims $\mathcal{P}(c_i)$, where $|\mathcal{P}(c_i)| \geq 0$. A claim with no parents is considered a root claim.

Events: An event is a coordination action that operates on claims or subtasks. Events define how claims are created, modified, and connected, and therefore determine the structure of coordination.

Each event is associated with:

- an event type (e.g., `delegate_subtask`, `revise_claim`, `contradict_claim`, `merge_claims`),
- an acting agent,
- a target claim or subtask,
- and a resulting claim (if applicable).

Event-Induced Claim Transitions: Each event type induces a specific transformation in the claim structure:

- **Revision:** creates a new claim with a single parent, forming a chain.
- **Contradiction:** creates a new claim that challenges an existing claim, introducing branching.
- **Merge:** creates a new claim with multiple parent claims, introducing multi-parent dependencies.
- **Delegation:** creates a new subtask and initiates new claims within that subtask.

These transitions define the edges of the claim graph and collectively produce a directed acyclic graph (DAG) over claims.

Key Property: The coordination structure is not imposed a priori but emerges from event-induced claim transitions. In particular, merge operations introduce multi-parent dependencies, converting tree-like reasoning structures into directed acyclic graphs.

F.3 Trace Logging Schema

We record fine-grained coordination traces at the event level to enable reconstruction of coordination structures. Each interaction step produces a structured record containing event metadata, claim lineage, subtask context, and derived coordination attributes.

Overview: Each logged entry corresponds to a single coordination event and includes identifiers that link it to claims, subtasks, and other events. These records collectively define the inputs required to reconstruct both the subtask tree and the claim DAG.

Event-Level Fields.

Claim Graph Fields: The `root_claim_id` is propagated across lineage and enables reconstruction of full coordination cascades.

Subtask Tree Fields.

Derived Coordination Fields. These fields are computed from raw interaction traces and are not directly provided by the system. They enable identification of revision waves, contradiction bursts, and merge operations during post-processing.

Claim Type	Description
Proposed Claim	An initial statement or solution generated by an agent.
Revised Claim	A modification of a prior claim that refines or corrects it.
Contradictory Claim	A claim that challenges or disputes an existing claim.
Merged Claim	A claim produced by combining multiple parent claims.

Table 15: Types of claims observed in coordination traces. Claims represent reasoning artifacts that are created and transformed through events.

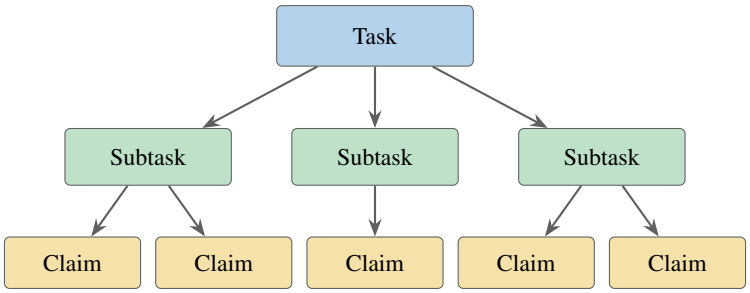


Figure 12: Hierarchy of coordination structures in a multi-agent system. A task defines the global objective, which is decomposed into subtasks; subtasks produce claims; and claims evolve through event-level coordination steps. Our analysis operates on these events rather than on whole tasks.

Reconstructability: Together, these fields provide sufficient information to reconstruct:

- the subtask tree from delegation events,
- the claim DAG from parent–child relationships,
- and coordination cascades from root claim identifiers.

F.4 DAG Construction Pipeline

We reconstruct coordination structures from event traces in two stages: (i) subtask-tree construction from delegation events, and (ii) claim-DAG construction from claim-level lineage fields. Figure 15 summarizes this pipeline.

Subtask Tree Construction. Delegation events define a hierarchical decomposition over work units. For each `delegate_subtask` event, we create a new subtask node and add a directed edge from `parent_subtask_id` to `subtask_id`. This produces a rooted tree over subtasks, where depth corresponds to recursive decomposition depth. The subtask tree therefore captures *what agents are asked to work on*.

Claim DAG Construction. Claims form a separate structure that captures *how reasoning evolves*. Each claim is represented as a node indexed by `claim_id`. Directed edges are created from every element of `parent_claim_ids` to the current claim. This yields three canonical cases:

- **Revision:** a claim has exactly one parent, producing a chain.
- **Contradiction:** multiple child claims may reference the same parent, producing branching.
- **Merge:** a claim has multiple parents, producing a multi-parent node.

Because merge operations introduce multiple incoming edges, the resulting structure is in general a directed acyclic graph (DAG), rather than a tree.

Root Claim Assignment. Each claim is associated with a `root_claim_id`, inherited from its earliest ancestor. Claims with no parents are initialized as root claims. For descendant claims, the root identifier is propagated through lineage during post-processing. This field is critical because it makes cascades reconstructable from local parent–child relationships.

Derived Coordination Grouping. Several higher-level coordination structures are not directly emitted by the system and must be derived from raw traces:

- **Revision chains** are formed by grouping claims linked through iterative single-parent revision steps.

Event Type	Operation on Claims	Resulting Structure
`revise_claim`	single parent → child	chain
`contradict_claim`	parent → multiple children	branching
`merge_claims`	multiple parents → child	multi-parent (DAG)
`delegate_subtask`	creates new subtask context	hierarchical tree

Table 16: Event types and their induced transformations on the claim structure. These operations define how the claim graph evolves from chains to branching structures and ultimately to a directed acyclic graph through merge operations.

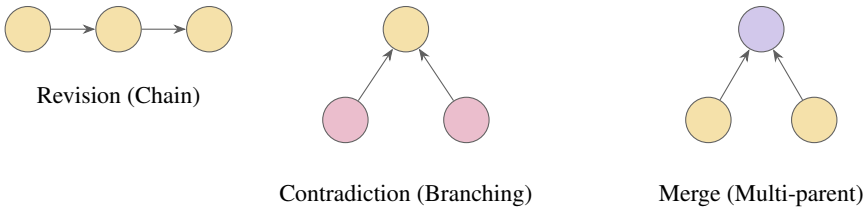


Figure 13: Event-induced transformations of the claim structure. Revision produces linear chains, contradiction introduces branching around a claim, and merge combines multiple parent claims into a single node, yielding a directed acyclic graph.

- **Contradiction groups** are formed by grouping claims that reference the same parent claim within a temporal window.
- **Merge groups** are formed by collecting all parent claims participating in a `merge_claims` event.

These derived identifiers are used to extract revision waves, contradiction bursts, and merge fan-in during analysis. The trace schema therefore records both raw lineage fields and post-processed grouping fields needed for event-level measurement.

Cascade Extraction. Given a root claim c_{root}, we define its coordination cascade as the reachable subgraph of all downstream claim and event instances associated with that root. Operationally, cascade extraction begins from `root_claim_id` and traverses all descendant claims connected through revision, contradiction, merge, and delegation-linked activity. This reachable subgraph defines the root-centered cascade over which aggregate observables are computed. In particular, TCE counts the downstream coordination events associated with the cascade rather than the number of claim nodes in the subgraph.

Workload DAG, Subtask Tree, and Claim DAG. These three structures serve distinct roles. The workload DAG is constructed before execution by the task-expansion module and specifies task dependencies without prescribing coordination. During execution, delegation events induce a runtime subtask tree that records recursive work decomposition. Separately, the claim DAG records the emergent propagation, revision, contradiction, and integration of reasoning across agents.

Reproducibility. This construction procedure ensures that all coordination observables reported in the paper are computed from structured traces rather than manually annotated or heuristically imposed interaction graphs. Given the logged fields in Section F.3, the subtask tree, claim DAG, and cascade assignments are fully reconstructable.

F.5 Worked Example: From Task Expansion to Coordination Cascade

We illustrate the full coordination pipeline using a representative configuration from our experimental setup. We consider a system with $N = 16$ agents operating on a GAIA-style multi-step reasoning task. As described in Section 3, each task is expanded into an interdependent workload DAG using a structured reasoning procedure..

Task Expansion. Given an input task requiring multi-step reasoning and constraint satisfaction, the expansion module generates the following interdependent task structure:

- **T0:** Solve the full problem
- **T1:** Extract key entities and constraints from the prompt
- **T2:** Generate candidate reasoning paths

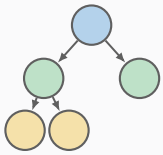
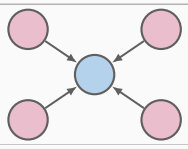
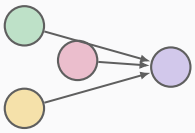
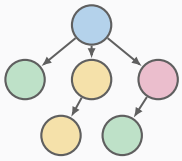
Event Type	Schematic	Definition	What It Captures
Delegation Cascade		Number of events in the subtask tree rooted at a `delegate_subtask` event.	Recursive task decomposition and agent recruitment.
Revision Wave		Length of a chain of `revise_claim` events linked by `parent_claim_id`.	Iterative refinement of a claim.
Contradiction Burst		Number of distinct agents issuing `contradict_claim` on the same parent claim.	Parallel critique centered on one claim.
Merge Fan-in		Number of `parent_claim_ids` referenced by a single `merge_claims` event.	Information integration bottleneck.
Total Cognitive Effort (TCE)		Total number of downstream coordination events linked to a root claim.	Aggregate cascade size.

Table 17: **Coordination-event types used in the our power-law analysis.** Primitive logged actions: delegation, revision, contradiction, and merge, are linked through claim and subtask relationships to construct the event types analyzed here: delegation cascades, revision waves, contradiction bursts, and merge fan-in. Total Cognitive Effort (TCE) is an aggregate root-cascade measure over downstream coordination events. **Legend:** *Root Claim* *Propagation Claims* *Critique Claim* *Merge Claim.*

Field	Description
run_id	Identifier for the experiment run
step_id	Temporal ordering of events
agent_id	Acting agent
event_type	Type of coordination event
target_claim_id	Referenced claim (if applicable)
target_subtask_id	Referenced subtask (if applicable)
timestamp	Event time
message_length	Token count (proxy for cost)

Table 18: Event-level fields recorded for each coordination step.

- **T3:** Evaluate candidates under extracted constraints
- **T4:** Refine top candidates
- **T5:** Integrate reasoning into final answer

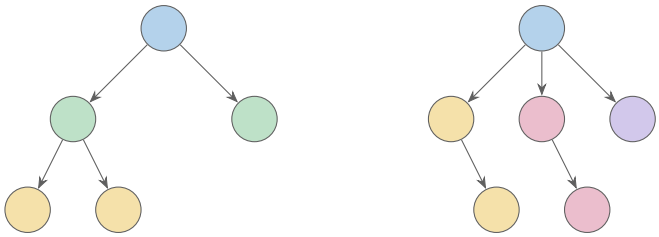
Dependencies are not purely hierarchical: T3 depends on both T1 and T2, T4 depends on T3, and T5 integrates outputs from T2 and T4. This produces a directed workload DAG with cross-subtask dependencies.

Workload DAG. Figure 16 (left) shows the resulting task structure.

Agent Execution and Event Trace. Agents are assigned to subtasks and generate claims through interaction. A representative trace excerpt is:

- `c1: propose_claim (T2)` — initial reasoning path
- `c2: revise_claim(c1) (T3)` — adjusted using constraints from T1
- `c3: contradict_claim(c1) (T3)` — alternative reasoning path
- `c4: revise_claim(c2) (T4)` — refined evaluation
- `c5: merge_claims(c2, c3, c4) (T5)` — integrated final reasoning

All claims inherit `root_claim_id = c1`, forming a single coordination cascade.



Delegation (Subtask Tree) Cascade (Total Cognitive Effort)

Figure 14: **Delegation and cascade structure.** Delegation events construct a subtask tree (left), defining how work is decomposed. Coordination cascades (right) collect the downstream coordination events associated with a root claim, including revision, contradiction, merge, and delegation-linked activity. Total Cognitive Effort (TCE) is the total number of downstream coordination events in the root-centered cascade.

Field	Description
claim_id	Unique identifier for each claim
parent_claim_ids	Parent claims defining DAG edges
root_claim_id	Root ancestor of the claim
claim_depth	Depth within the claim DAG
claim_status	Proposed, revised, contradictory, or merged

Table 19: Fields defining the claim-level DAG structure.

Field	Description
subtask_id	Unique identifier for each subtask
parent_subtask_id	Parent subtask in hierarchy
subtask_depth	Depth in subtask tree
assigned_agent	Responsible agent
subtask_status	Active or completed

Table 20: Fields defining the hierarchical subtask decomposition.

Coordination Structures. Figure 16 shows the full pipeline: (left) task expansion into an interdependent structure, (center) claim-level DAG constructed from event traces, and (right) the cascade extracted from the root claim.

Observable Computation. From the cascade rooted at c_1, we compute:

- **Revision Wave:** two linked `revise_claim` events, $c_1 \rightarrow c_2$ and $c_2 \rightarrow c_4$ (wave length $= 2$).
- **Contradiction Burst:** one agent issues a `contradict_claim` event targeting c_1, producing c_3 (burst size $= 1$).
- **Merge Fan-in:** c_5 integrates c_2, c_3, and c_4 (fan-in $= 3$).
- **Total Cognitive Effort (TCE):** four downstream coordination events are associated with the cascade rooted at c_1: two revisions, one contradiction, and one merge (event count $= 4$).

Interpretation. This example reflects the coordination dynamics induced by our workload expansion procedure: task decomposition introduces interdependent subtasks, agents generate competing and refined reasoning across these subtasks, and merge operations integrate multiple branches. The resulting cascade structure is the fundamental object used to study scaling laws of coordination in our experiments.

F.6 Observable Trigger Conditions

We define the exact extraction rules used to compute coordination observables from reconstructed structures. All observables are computed from either the claim DAG, the subtask tree, or root-centered cascades as described in Sections F.4 and F.5.

Notes. All grouping variables (e.g., `revision_chain_id`, `contradiction_group_id`) are derived from raw traces during post-processing and are not directly emitted by the system. Temporal windows for contradiction grouping are defined relative to event timestamps. Observables are computed consistently across all tasks, topologies, and

Field	Description
revision_chain_id	Groups claims in a revision sequence
contradiction_group_id	Groups claims targeting the same parent claim
merge_id	Identifier for merge operations
merge_parent_ids	Parent claims involved in merge

Table 21: Derived fields used to group events into coordination structures.

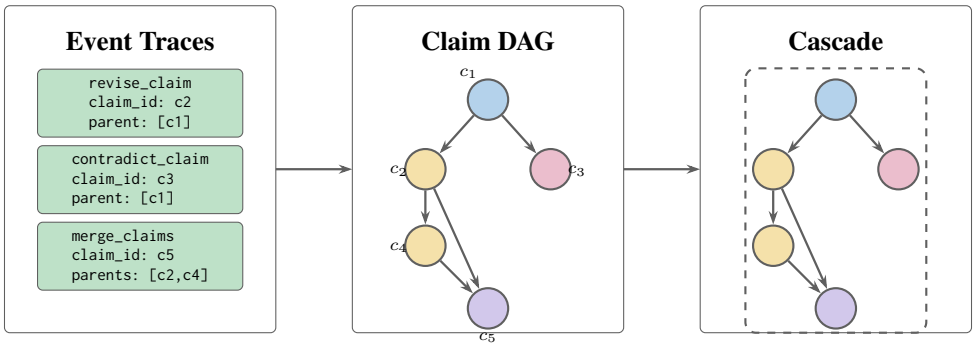


Figure 15: From structured event traces to coordination cascades. Logged event fields define parent–child claim relationships, from which we reconstruct the claim DAG. Cascades are then extracted as root-centered reachable subgraphs, enabling computation of event-level observables such as revision waves, contradiction bursts, merge fan-in, and total cognitive effort.

system scales.

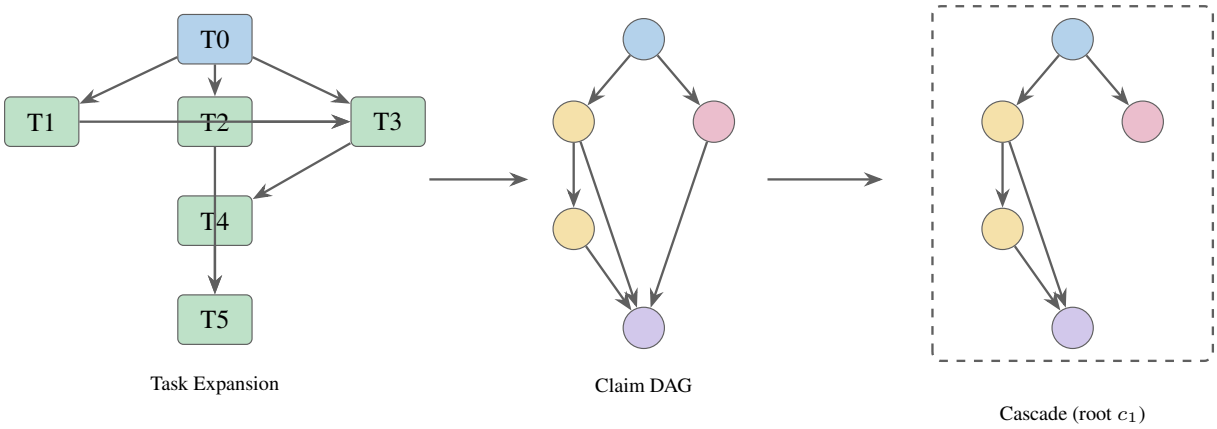


Figure 16: End-to-end coordination pipeline in our experimental setup. Left: task expansion produces an interdependent workload DAG with cross-subtask dependencies. Center: agent interactions generate claims and event-induced edges, forming a claim DAG. Right: the cascade extracted from the root claim captures the full coordination effort used to compute event-level observables.

Observable	**Trigger Condition**	**Structure**	**Output**
Delegation Cascade	Delegation subtree rooted at a `delegate_subtask` event	Subtask tree	Delegated-subtask count
Revision Wave	Maximal chain of linked `revise_claim` events	Claim DAG	Revision-event count
Contradiction Burst	Distinct agents issuing `contradict_claim` against the same parent claim within temporal window τ	Claim DAG + timestamps	Distinct-agent count
Merge Fan-in	Number of `parent_claim_ids` referenced by one `merge_claims` event	Claim DAG	Parent count
Total Cognitive Effort (TCE)	Downstream coordination events associated with `root_claim_id`	Cascade	Event count

Table 22: Formal trigger conditions for coordination observables. Each observable is computed from structured traces via deterministic extraction rules applied to reconstructed coordination structures.

G LLM Ablation

	Tail structure			LR tests		Pref. Reinf.	Concentration		Outcome
Model	$\hat{\alpha}$ [95% CI]	$\hat{x}_c$	$x_{\max}$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	$\hat{\beta}$	E_{10}^{all}	Active	Task Succ.
GPT-4o-mini	2.22 [2.17, 2.26]	57	1320	+5.2	+2.8	0.16	25%	81%	0.48
Qwen 2.5 72B	2.27 [2.18, 2.33]	44	1035	+5.0	+2.7	0.15	23%	79%	0.43
Llama 3.1 70B	2.34 [2.27, 2.39]	30	1012	+4.8	+2.5	0.13	19%	78%	0.41
Qwen 2.5 7B	2.74 [2.66, 2.81]	9	440	+3.5	+1.7	0.07	14%	72%	0.27

Table 23: **Global pooled TCE robustness across models.** All 16 topology×task conditions are pooled per model. Each model is fit independently using discrete MLE with a KS-selected $x_{\min}$ and run-clustered bootstrap inference. Tail statistics summarize the fitted truncated-power-law structure of Total Cognitive Effort (TCE), together with comparisons against log-normal and pure power-law alternatives. Preferential reinforcement is summarized by $\hat{\beta}$, concentration by the top-10% effort share and active-agent fraction, and outcome by overall task success. For the non-nested TPL–LN comparison, $R_{\mathrm{TPL/LN}} > 0$ favors TPL; for the nested TPL-PL comparison, $2\Delta\ell_{\mathrm{TPL/PL}}$ is reported with boundary-aware inference as described in Sec. 3.3.

		Tail structure		LR tests		Concentration	Outcome
Topology	**Model**	$\hat{\alpha}$ [95% CI]	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	E_{10}^{all}	Task succ.
Chain	GPT-4o-mini	2.46 [2.41, 2.48]	21	+4.5	+2.3	15%	0.43
	Qwen 2.5 72B	2.54 [2.47, 2.55]	23	+4.2	+2.7	14%	0.40
	Llama 3.1 70B	2.63 [2.55, 2.66]	17	+4.0	+2.2	13%	0.36
	Qwen 2.5 7B	2.78 [2.73, 2.81]	8	+3.1	+1.7	13%	0.19
Star	GPT-4o-mini	2.26 [2.19, 2.31]	42	+5.4	+2.6	32%	0.47
	Qwen 2.5 72B	2.23 [2.12, 2.28]	41	+5.4	+2.3	29%	0.44
	Llama 3.1 70B	2.29 [2.19, 2.33]	35	+4.6	+2.3	27%	0.42
	Qwen 2.5 7B	2.59 [2.48, 2.66]	11	+3.3	+1.8	19%	0.26
Hierarchical	GPT-4o-mini	2.21 [2.15, 2.29]	31	+5.3	+2.7	21%	0.53
	Qwen 2.5 72B	2.17 [2.08, 2.23]	28	+5.3	+2.4	18%	0.49
	Llama 3.1 70B	2.31 [2.24, 2.36]	28	+4.6	+2.8	15%	0.46
	Qwen 2.5 7B	2.63 [2.55, 2.67]	9	+3.4	+1.6	12%	0.31
Fully Connected	GPT-4o-mini	2.17 [2.13, 2.26]	61	+5.5	+3.0	32%	0.48
	Qwen 2.5 72B	2.16 [2.04, 2.21]	53	+5.9	+3.4	29%	0.46
	Llama 3.1 70B	2.22 [2.17, 2.31]	48	+5.4	+2.8	28%	0.42
	Qwen 2.5 7B	2.54 [2.44, 2.58]	15	+4.2	+2.6	15%	0.29

Table 24: **TCE heavy-tail structure by topology and model.** Results are pooled across task types, seeds, and agent-society sizes within each topology×model condition. Each condition is fit independently using discrete MLE with a KS-selected $x_{\min}$; uncertainty is estimated with run-clustered bootstrap resampling. Tail statistics report $\hat{\alpha}$ and $\hat{x}_c$ together with comparisons against log-normal and pure power-law alternatives. Concentration is summarized by E_{10}^{all} and outcome by pooled task success.

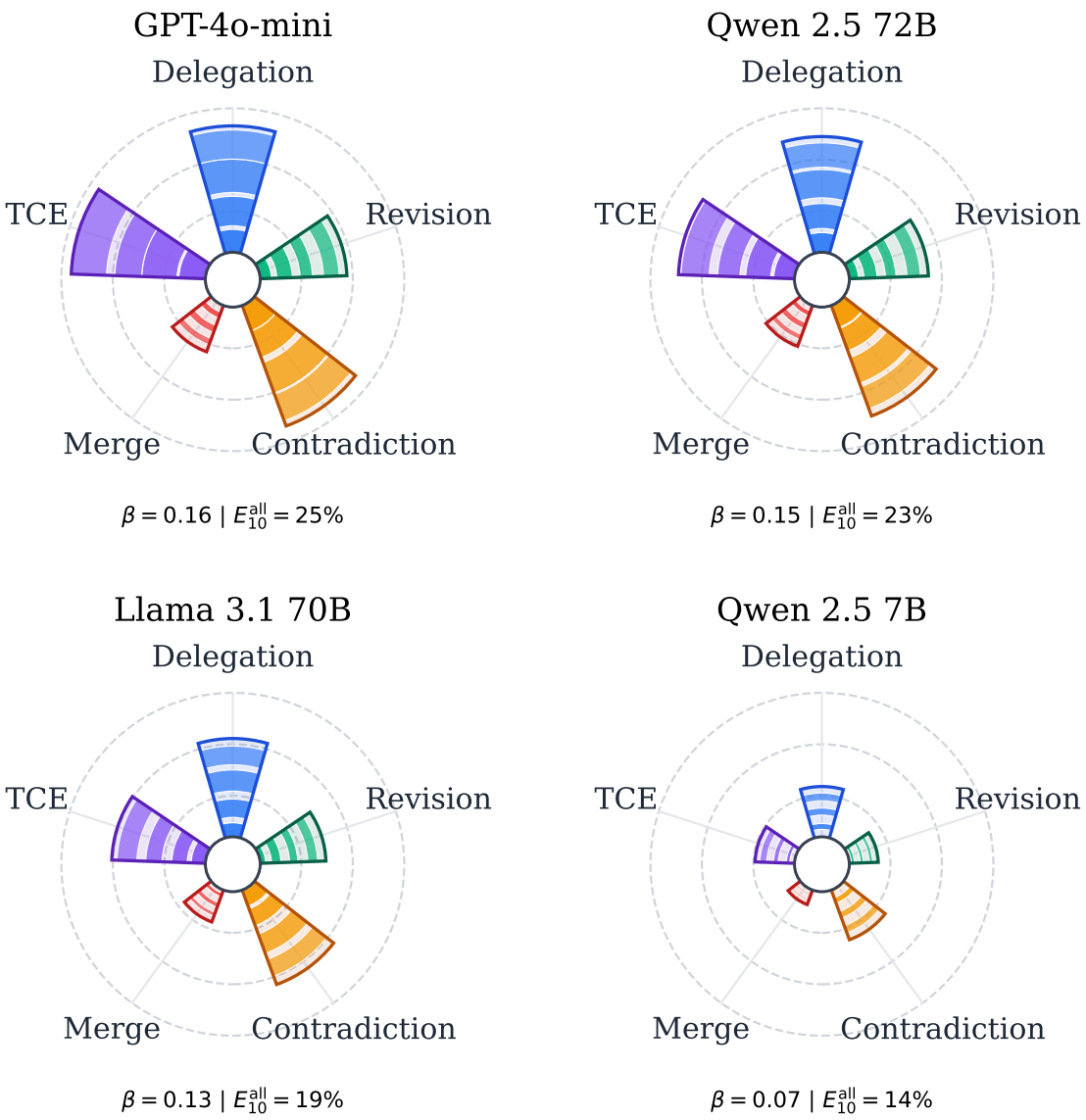


Figure 17: **Coordination-law flower signatures across models.** Each panel summarizes one model using five global event observables: delegation cascade, revision wave, contradiction burst, merge fan-in, and Total Cognitive Effort (TCE). Petal extent jointly reflects four law dimensions: heavier tails (lower $\hat{\alpha}$), larger truncation scale $\hat{x}_c$, stronger preferential reinforcement $\hat{\beta}$, and greater influence concentration E_{10}^{all}. Across all four models, the qualitative ordering is preserved: TCE remains the strongest composite coordination signal, delegation and contradiction form the next tier of broad coordination, revision is intermediate, and merge is the weakest and most localized. Across evaluated models, the signatures vary in magnitude while retaining this overall structure, with more contracted profiles exhibiting shorter fitted tails, lower cutoff scales, and weaker measured influence concentration.

		Tail structure		LR tests		Concentration	Outcome
Task type	**Model**	$\hat{\alpha}$ [95% CI]	$\hat{x}_c$	$R_{\mathrm{TPL/LN}}$	$2\Delta\ell_{\mathrm{TPL/PL}}$	E_{10}^{all}	Task succ.
Planning	GPT-4o-mini	2.14 [2.03, 2.19]	52	+5.1	+2.3	30%	0.45
	Qwen 2.5 72B	2.17 [2.16, 2.27]	48	+5.3	+2.1	27%	0.41
	Llama 3.1 70B	2.23 [2.11, 2.30]	45	+4.6	+1.7	20%	0.40
	Qwen 2.5 7B	2.49 [2.39, 2.53]	15	+3.7	+1.3	18%	0.23
Reasoning	GPT-4o-mini	2.26 [2.15, 2.32]	48	+5.5	+2.4	23%	0.56
	Qwen 2.5 72B	2.25 [2.18, 2.27]	40	+4.9	+2.5	23%	0.52
	Llama 3.1 70B	2.35 [2.22, 2.45]	37	+4.6	+2.7	19%	0.51
	Qwen 2.5 7B	2.66 [2.52, 2.71]	13	+3.4	+1.7	14%	0.34
Coding	GPT-4o-mini	2.29 [2.17, 2.36]	34	+5.1	+3.2	26%	0.46
	Qwen 2.5 72B	2.37 [2.28, 2.42]	34	+5.3	+2.6	21%	0.44
	Llama 3.1 70B	2.44 [2.35, 2.51]	27	+4.8	+2.5	21%	0.42
	Qwen 2.5 7B	2.73 [2.59, 2.78]	9	+3.5	+1.9	13%	0.29
QA	GPT-4o-mini	2.38 [2.24, 2.47]	27	+5.6	+3.5	21%	0.51
	Qwen 2.5 72B	2.47 [2.39, 2.50]	23	+5.0	+3.6	19%	0.47
	Llama 3.1 70B	2.53 [2.42, 2.56]	20	+4.6	+3.6	18%	0.44
	Qwen 2.5 7B	2.81 [2.66, 2.84]	9	+3.5	+2.3	15%	0.33

Table 25: **TCE heavy-tail structure by task type and model.** Results are pooled across topologies, seeds, and agent-society sizes within each task-type×model condition. Each condition is fit independently using discrete MLE with a KS-selected $x_{\min}$, with uncertainty estimated by run-clustered bootstrap resampling. Tail statistics report the fitted truncated-power-law parameters for Total Cognitive Effort (TCE), together with model-comparison statistics against log-normal and pure power-law alternatives. Concentration is summarized by the top-10% effort share E_{10}^{all}, and outcome by pooled task success.

H Additional Details on Experimental Setup

This section summarizes the experimental configurations, benchmark coverage, agent scales, and coordination-event sample counts used throughout the study. Additional details on workload construction and agent configuration are provided in Appendices I and J.

Component	Configuration
Benchmarks	GAIA, SWE-bench Verified, REALM-Bench, and MultiAgentBench; benchmark and workload accounting is reported in Table 12
Task types	QA, reasoning, coding, planning
Agent counts (N)	$\{8, 16, 32, 64, 128, 256, 512\}$
Topologies	Chain, Star, Tree, Hierarchical, Fully Connected, Sparse Mesh, Dynamic Reputation
Primary agent model	Shared LLM (GPT-4o-mini)
Execution horizon	20 steps per workflow
Random execution seeds	5 independent repetitions per experimental configuration
Task expansion	Benchmark-conditioned, size-dependent workload expansion (Appendix I)
Trace granularity	Event-level (delegation, revision, contradiction, merge)
Completed workflows	$\sim$ 98,000

Table 26: **Experimental configuration summary.** Expansion seed tasks used for workload generation are distinct from the five random execution seeds used to repeat each configuration. The reported workflow count is the aggregate number of completed LangGraph executions, rather than a product of a single fixed task count.

Observable	Samples	Approx. mean/run
Delegation cascades	342,323	~ 3.49
Revision waves	344,356	~ 3.51
Contradiction bursts	255,475	~ 2.61
Merge events	278,533	~ 2.84
TCE (root cascades)	411,370	~ 4.20
Total	**1,632,057**	$\sim$ **16.65**

Table 27: Coordination-event samples extracted from approximately 98,000 runs. Sample counts are reported exactly; mean/run values are approximate because the run count is rounded for presentation. Each run may produce multiple event instances depending on the realized interaction trace.

I Workload Expansion Module

To study coordination across agent societies of varying size, we use a benchmark-conditioned workload expansion module that scales the execution workload with N while preserving task diversity and executability. The purpose of this expansion is to maintain meaningful agent utilization as society size increases, rather than allowing large-N systems to become artificially underloaded.

The module is explicitly constrained to generate *only workload* and does not prescribe any coordination structure. Tasks and their dependencies are constructed before execution and independently of the communication topology. Delegation, revision, contradiction, and merge events arise only from the realized interaction traces, from which aggregate Total Cognitive Effort (TCE) is subsequently computed.

For each benchmark b and task family d, we sample $K = \min(5, |B(b, d)|)$ seed tasks that provide benchmark-grounded problem contexts for workload generation. A generator LLM produces related candidate tasks from these contexts, which are filtered by a validator using criteria of same-world consistency, non-paraphrase, independent meaningfulness, executability, and additive informativeness. The resulting validated tasks form the benchmark-conditioned pool from which execution workloads are constructed.

For a society of size N, the execution workload contains

$$T(N) = \lceil N^{0.65} \rceil$$

subtasks. This induces the size-dependent agents-per-subtask ratio

$$\rho(N) = \frac{N}{T(N)} = \frac{N}{\lceil N^{0.65} \rceil}.$$

Thus, workload grows sublinearly with society size: larger societies receive more executable subtasks while also allowing progressively more agents to participate in each subtask. The selected subtasks are connected via sparse, randomly sampled dependency edges, forming a shallow workload DAG independent of the communication topology.

Figure 18 verifies the resulting operating regime empirically. Active-agent participation remains high across society sizes, while agents per subtask increase gradually with N according to the size-dependent workload rule. The main scaling sweep therefore studies larger societies under increasing but sublinear workload, preserving broad participation while allowing their realized coordination dynamics to emerge.

I.1 Fixed-Workload Validation

We complement the utilization-preserving scaling study with a fixed-workload validation that char-

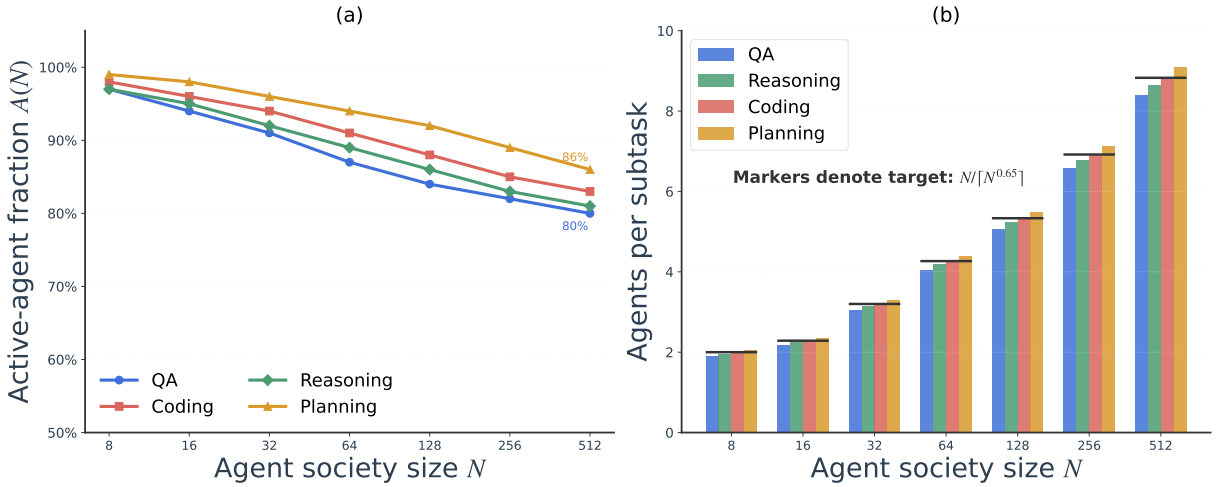


Figure 18: **Workload-expansion validation across agent society size. (a)** Active-agent fraction $A(N)$ for four task families remains high across scales, staying above 80% even at $N = 512$, indicating sustained participation as society size grows. **(b)** Agents per subtask induced by the size-dependent workload rule, shown against the target $\rho(N) = N/\lceil N^{0.65} \rceil$. The gradual increase in agents per subtask accompanies a growing workload while avoiding concentration onto a small fixed task set. Together, these results show that the expansion procedure maintains broad agent utilization across scales without prescribing the coordination behavior analyzed in the main experiments.

acterizes scaling with society size under an unchanged task load. In the Planning×Mesh condition, using the same five seeds, we compare the standard $N = 64$ setting with workload W_{64}, the standard $N = 128$ setting with workload W_{128}, and a fixed-workload condition in which society size is increased to $N = 128$ while retaining the exact W_{64} workload.

Maximum TCE increases substantially under fixed workload, from 180 to 285, retaining approximately 66% of the corresponding increase observed in the standard $N = 64 \rightarrow 128$ scaling condition. The fitted exponents remain stable within uncertainty. At the same time, the larger increase in $\hat{x}_c$ under W_{128} shows that workload expansion further extends the accessible tail range in this setting.

Together, the two protocols provide complementary views of scaling. The main sweep characterizes coordination in broadly utilized societies under size-dependent workload growth, while the fixed-workload validation shows that the same extreme-event growth pattern persists when task load is held constant.

Condition	N	Workload	TCE $x_{\max}$	$\hat{\alpha}$ [95% CI]	$\hat{x}_c$ [95% CI]
Standard	64	W_{64}	180 ± 20	2.25 [2.09, 2.42]	190 [173, 237]
Fixed workload	128	W_{64}	285 ± 36	2.19 [2.14, 2.40]	215 [190, 239]
Standard	128	W_{128}	340 ± 41	2.17 [2.04, 2.31]	322 [298, 415]

Table 28: **Extreme-event scaling under fixed workload.** In Planning×Mesh, increasing society size from $N = 64$ to $N = 128$ while holding the workload fixed at W_{64} increases maximum TCE from 180 ± 20 to 285 ± 36, preserving approximately 66% of the increase observed under the utilization-preserving workload-scaling protocol ($180 \pm 20 \rightarrow 340 \pm 41$). The fitted tail exponents remain similar with overlapping confidence intervals, while the cutoff changes only modestly under fixed workload and increases further under W_{128}.

J Agent Configuration and Experimental Protocol

All agents share a standardized configuration across all topologies, task families, scales, and seeds. No agent-level hyperparameters are tuned per condition, ensuring that observed differences arise solely from interaction structure and workload.

J.1 Agent Configuration

Component	Configuration
Orchestration	LangGraph (uniform orchestration, state, and execution)
Base prompt	Task, neighbors, structured history
Topology addendum	Topology-specific routing behavior (chain, star, tree, hierarchical, mesh, reputation)
Task addendum	Benchmark-specific instructions (QA, coding, planning, reasoning)
Tool access	Restricted to benchmark-native tools
Context budget	4000-token window
Completion budget	1000 tokens
Memory	No cross-run persistence; structured state only
Seeds	5 per configuration

Table 29: Standardized agent configuration used across all experiments.

J.2 Benchmarks

Benchmark	Task family	Primary capability
GAIA	QA / multimodal reasoning	Open-ended reasoning and tool use
SWE-bench Verified	Coding	Software debugging and repair
REALM-Bench	Planning / constraint reasoning	Dependency-aware planning and scheduling
MultiAgentBench	Coordination / interaction	Multi-agent collaboration and competition

Table 30: **Benchmark coverage.** Quantitative source-pool and workload-expansion accounting is reported in Table 12.

J.3 Prompt Structure

Each agent prompt consists of three layers: (i) a shared base prompt, (ii) a topology-specific addendum, and (iii) a task-family-specific addendum. The base prompt is identical across all agents and does not prescribe any coordination strategy.

Base prompt.

> You are an AI agent participating in a multi-agent reasoning system. Your goal is to contribute high-quality reasoning to solve the task.
>
> Core behaviors: - Think step by step before answering. - Identify and correct errors in other agents' outputs when necessary. - Build on useful prior reasoning rather than repeating it. - Be concise but complete.
>
> Output format: Provide reasoning followed by a final answer labeled: `ANSWER: <your answer>`

This prompt is intentionally minimal and strategy-agnostic. No explicit instructions for delegation, critique, or merging are included, ensuring that coordination patterns emerge from interaction rather than prompt design.

Example topology addendum (reputation routing).

> You are agent {agent_id} in a reputation-routed network (step {step} of {max_steps}). You have consulted a set of peers selected by reputation.
>
> - Review consulted outputs critically. - Do not blindly trust high-reputation agents. - Revise your reasoning using useful information. - Prefer correction and synthesis over agreement.
>
> At intermediate steps, produce an improved claim. At the final step, synthesize the best answer.

Topology-specific addenda define communication and routing behavior but do not impose specific coordination patterns. Task-specific addenda provide domain context (e.g., QA, coding, planning) while maintaining the same interaction structure.

Overall, this design ensures that coordination dynamics arise from agent interaction under shared constraints, rather than from prompt engineering or task-specific scripting.